



MASTER'S THESIS

Efficient Portfolios under convex risk measures

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Abstract

The classical mean-variance approach to portfolio choice treats variance as a synonym for risk, an identification that fails on two counts : variance penalises upside and downside deviations identically, and it is not monotone, so a position that dominates another in every state of the world can carry higher variance. This thesis develops, from first principles, the axiomatic theory of convex and coherent risk measures that resolves this difficulty, and formulates and solves the resulting efficient portfolio problem across the mono-period, discrete-time multi-period and continuous-time settings, following Föllmer and Schied’s convex risk measure framework, Pliska’s discrete-time market model, and Taflin’s insurance application, complemented throughout by numerical simulations.

The robust representation theorem for convex risk measures is proved in full for finite probability spaces ; Value-at-Risk’s failure of subadditivity is exhibited by an explicit, parametrically verified counterexample ; and Average Value-at-Risk is adopted as the coherent, numerically tractable risk measure used throughout. In the mono-period setting, the AVaR-efficient portfolio problem is shown to coincide exactly with the classical Markowitz problem under elliptical returns, and to diverge from it, substantially, once returns are skewed. In discrete time, a recursive composition of one-step conditional risk assessments is proved to be automatically time-consistent, and a random search over scenario trees exhibits genuine, large divergence from the naive alternative that recomputes the risk measure directly on the terminal law. In continuous time, the AVaR-optimal constant-mix proportion is derived in closed form as the classical Merton myopic proportion minus an explicit risk-aversion correction. All of these results are implemented and tested in an accompanying Python package (45 tests, five executed notebooks), which also contributes, beyond the scope of the required references, a nested-bootstrap correction for the estimation risk inherent in optimising a risk measure against a finite sample, verified directly against the axioms of a convex risk measure, and found to be well-calibrated but, honestly reported, not itself coherent.

Keywords : convex risk measures, coherent risk measures, Average Value-at-Risk, Value-at-Risk, efficient portfolios, Markowitz, time-consistency, dynamic risk measures, distributionally robust optimisation, optimizer's curse.

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Chapitre 1

Introduction

The classical starting point of portfolio theory is Markowitz's mean-variance framework (1), in which an investor chooses among risky assets so as to maximise expected return for a given level of variance, or equivalently minimise variance for a given expected return. This formulation gave portfolio choice its first rigorous mathematical foundation and remains, to this day, the reference model taught to every practitioner. Its elegance, however, rests on treating variance as a synonym for risk, an identification that does not survive close scrutiny. Variance penalises upside and downside deviations from the mean identically, whereas an investor's genuine concern is asymmetric : a large unexpected gain is not a risk in the same sense as a large unexpected loss. More fundamentally, variance is not *monotone* : a position that dominates another in every state of the world, and is therefore unambiguously preferable, can nonetheless carry a higher variance. A measure of risk that can penalise a strictly better position more heavily than a strictly worse one is, on reflection, not measuring risk at all.

This difficulty motivated the search for an axiomatic theory of risk measurement, one that specifies, from first principles, the properties a quantity must satisfy before it can legitimately be called a measure of risk. A first answer, and the one still most widely used in practice, is the Value-at-Risk (VaR) : the loss that will not be exceeded with a given confidence level over a given horizon. VaR is intuitive and easy to communicate, which explains its adoption by regulators, but it suffers from a defect that is more than a technical nuisance : it is not, in general, subadditive. The risk of a merged portfolio, as measured by VaR, can exceed the sum of the risks of its constituent parts, so that VaR can *penalise* diversification, the single most basic risk-management principle. This failure is not a pathological edge case ; it occurs precisely in the presence of the skewed, fat-tailed

dependence structures that are most characteristic of real financial returns, and that a sound measure of risk should be best equipped to capture.

The response to this difficulty, and the theoretical backbone of this thesis, is the axiomatic theory of coherent and convex risk measures. Artzner, Delbaen, Eber and Heath (2) first proposed a small set of axioms monotonicity, translation invariance, positive homogeneity, and subadditivity that any *coherent* measure of risk should satisfy, ruling out VaR's failure of diversification benefit by construction. Föllmer and Schied (3) subsequently observed that positive homogeneity and subadditivity, while natural in frictionless markets, become questionable once trading itself is subject to constraints or liquidity costs : doubling a position need not double its risk if it cannot be doubled without moving the market. Relaxing these two axioms into the single, weaker requirement of *convexity* yields the broader class of *convex* risk measures, together with a robust representation theorem : every such measure can be written as a worst-case expected loss over a family of probabilistic scenarios, penalised by a function that quantifies how much each scenario departs from a reference model. This representation is the analytical tool around which the present thesis is built, and it is developed in full in Föllmer and Schied's textbook treatment (4).

Once risk is measured this way, the natural counterpart to Markowitz's mean-variance frontier is an *efficient frontier under a convex risk measure constraint* : portfolios that maximise expected return subject to a bound on their convex risk, or dually, that minimise convex risk subject to a target return. This thesis develops this problem in three settings of increasing generality. In the mono-period setting, the problem reduces to a static optimisation exploiting the robust representation directly. In the discrete-time, multi-period setting, following Pliska's dynamic-programming framework (5), the problem must be solved by backward induction on conditional risk measures, raising the question of *time-consistency* : whether a strategy deemed optimal today remains optimal when re-evaluated at a later date, a question with no counterpart in the mono-period case. An insurance application of this multi-period framework, in which asset allocation is constrained by liabilities and regulatory requirements, is developed following Taflin (6). The continuous-time setting extends the analysis to dynamic risk measures evolving continuously through time.

Alongside this theoretical development, the thesis addresses the numerical estimation of the quantities that convex risk measures require in practice, in particular the reliable estimation of extreme quantiles from finite samples, following Bassi, Embrechts and Ka-

fetzaki (7), and situates this axiomatic approach against the older, more heuristic practice of downside-risk portfolio construction developed by Harlow (8), which anticipated some of the same asymmetric concerns without the full apparatus of a robust representation theorem. Numerical simulations are presented throughout to illustrate and validate the theoretical constructions.

The thesis is organised as follows. It first reviews the axiomatic theory of coherent and convex risk measures and situates it against Value-at-Risk and downside-risk approaches. It then develops the optimal portfolio problem under a convex risk measure constraint in the mono-period setting, before extending the analysis to the discrete-time multi-period setting and its insurance application, and to the continuous-time setting. Numerical methods and simulations are presented throughout, and the thesis closes with a discussion of the limitations of the framework and possible extensions.

Chapitre 2

Litterature Review

This chapter develops the axiomatic theory of monetary risk measures that underlies the whole thesis. Sections 2.1, 2.2 follow Föllmer and Schied (3; 4); Section 2.3 situates Value-at-Risk against this framework; Section 2.4 develops Average Value-at-Risk as the coherent alternative that this thesis uses throughout; Sections 2.5 and 2.6 close the chapter with the two remaining required references, quantile estimation and downside-risk portfolio construction, and position them relative to the axiomatic framework.

2.1 Financial positions and monetary risk measures

Fix a measurable space $(\Omega, \mathcal{F})$ and a set $\mathcal{X}$ of bounded, $\mathcal{F}$ -measurable functions $X : \Omega \rightarrow R$, closed under addition of constants, representing the discounted net worth of a financial position at a fixed future date. A *monetary risk measure* is a functional $\rho : \mathcal{X} \rightarrow R$ satisfying :

- (i) **Monotonicity.** If $X \leq Y$ pointwise on Ω , then $\rho(X) \geq \rho(Y)$: a position that dominates another in every state of the world cannot be assigned a higher capital requirement.
- (ii) **Cash invariance.** For every $m \in R$, $\rho(X+m) = \rho(X) - m$: adding a sure amount m to the position reduces the required capital by exactly m .

Together, (i) and (ii) already rule out the degenerate case : they imply ρ is Lipschitz continuous with respect to the supremum norm, since $X \leq Y + \|X - Y\|_\infty$ gives $\rho(X) \geq \rho(Y) - \|X - Y\|_\infty$ by monotonicity and cash invariance, and symmetrically for the reverse inequality. A risk measure is *normalised* if $\rho(0) = 0$, which we assume throughout without loss of generality (replace ρ by $\rho - \rho(0)$).

A monetary risk measure ρ is *convex* if

$$\rho(\lambda X + (1 - \lambda)Y) \leq \lambda\rho(X) + (1 - \lambda)\rho(Y), \quad \forall X, Y \in \mathcal{X}, \lambda \in [0, 1]. \quad (2.1)$$

Convexity formalises a diversification principle at the level of capital requirements : mixing two positions cannot increase the required capital beyond the corresponding mixture of their individual requirements. This is precisely the property that fails for Value-at-Risk, as shown in Section 2.3.

A convex risk measure ρ is *coherent* if it is in addition *positively homogeneous* :

$$\rho(\lambda X) = \lambda\rho(X), \quad \forall X \in \mathcal{X}, \lambda \geq 0. \quad (2.2)$$

Positive homogeneity together with convexity implies *subadditivity*,

$$\rho(X + Y) \leq \rho(X) + \rho(Y), \quad (2.3)$$

obtained from (2.1) with $\lambda = \frac{1}{2}$ and then rescaling both sides by 2 using (2.2). This recovers the original four-axiom definition of *coherent* risk measures introduced by Artzner, Delbaen, Eber and Heath (2) (monotonicity, cash invariance, positive homogeneity, subadditivity) : coherence is the special case of convexity in which risk scales linearly with position size.

Föllmer and Schied (3) observe that positive homogeneity is a strong and, in the presence of trading constraints or liquidity effects, an economically questionable requirement : doubling a position in an illiquid or constrained market does not, in general, double its risk, since unwinding a larger position moves prices against the holder. Subadditivity itself is a consequence of positive homogeneity combined with convexity rather than an independent economic requirement in this more general setting. Dropping positive homogeneity and with it subadditivity as a separate axiom, retaining only the weaker inequality (2.1) yields the strictly larger class of convex risk measures used throughout this thesis. Every coherent risk measure is convex ; the converse is false, as Example 2.1.1 below illustrates.

2.1.1 Acceptance sets

Every monetary risk measure is characterised by the set of positions it deems acceptable without any additional capital.

The *acceptance set* of a risk measure ρ is

$$\mathcal{A}_\rho = \{X \in \mathcal{X} : \rho(X) \leq 0\}. \quad (2.4)$$

Conversely, any set $\mathcal{A} \subseteq \mathcal{X}$ satisfying a mild monotonicity condition (if $X \in \mathcal{A}$ and $Y \geq X$ then $Y \in \mathcal{A}$) and $\inf\{m : m \in \mathcal{A}\} > -\infty$ generates a monetary risk measure by

$$\rho_{\mathcal{A}}(X) = \inf\{m \in R : m + X \in \mathcal{A}\}, \quad (2.5)$$

the smallest capital injection that renders X acceptable. The two constructions are mutually inverse : $\rho_{\mathcal{A}_\rho} = \rho$, and $\mathcal{A}_{\rho_{\mathcal{A}}}$ is the smallest acceptance set containing $\mathcal{A}$ that is closed under the cash-invariance relation. Moreover, ρ is convex if and only if $\mathcal{A}_\rho$ is a convex set, and ρ is coherent if and only if $\mathcal{A}_\rho$ is a convex cone. This equivalence is the basis of the robust representation theorem of Section 2.2 : characterising convex risk measures reduces to characterising convex acceptance sets.

Let Q be a finite family of probability measures on $(\Omega, \mathcal{F})$, each absolutely continuous with respect to a reference measure P , and let $\gamma : Q \rightarrow [0, \infty)$ be any function with $\min_{Q \in Q} \gamma(Q) = 0$. Define

$$\rho(X) = \max_{Q \in Q} (E_Q[-X] - \gamma(Q)).$$

This is a maximum of affine (hence convex) functionals of X , so ρ is convex ; it is monotone and cash-invariant because each term $E_Q[-X] - \gamma(Q)$ is. It fails positive homogeneity whenever γ is not identically zero, since scaling X by λ scales the $E_Q[-X]$ term by λ but leaves $\gamma(Q)$ unchanged. This is exactly the structure recovered, in full generality, by the representation theorem below : a convex risk measure is a worst-case expected loss over a family of scenarios, penalised by a function that need not vanish identically (coherence is precisely the degenerate case where the penalty only takes the values 0 and $+\infty$).

2.2 Robust representation of convex risk measures

The central analytical result of this chapter is that every convex risk measure with suitable regularity admits a dual description as a worst-case expected loss over a family of probabilistic models, penalised for departing from a reference view. We give the argument in full for Ω finite, where the proof is an elementary application of convex duality ;

the extension to general Ω requires a continuity property (the Fatou property, or lower semicontinuity with respect to bounded pointwise convergence) and is developed in full generality by Föllmer and Schied (4).

Let Ω be finite and let $\rho : \mathcal{X} \rightarrow R$ be a convex risk measure, where $\mathcal{X}$ is the (finite-dimensional) space of all functions on Ω . Then

$$\rho(X) = \max_{Q \in \mathcal{M}_1(\Omega)} (E_Q[-X] - \alpha_{\min}(Q)), \quad (2.6)$$

where $\mathcal{M}_1(\Omega)$ is the set of all probability measures on Ω and $\alpha_{\min}$, the *minimal penalty function*, is given by

$$\alpha_{\min}(Q) = \sup_{X \in \mathcal{A}_\rho} E_Q[-X]. \quad (2.7)$$

Moreover ρ is coherent if and only if $\alpha_{\min}$ takes values in $\{0, \infty\}$, i.e. $\alpha_{\min} = \iota_{\mathcal{Q}}$ (the indicator function, in the convex-analysis sense, of a set $\mathcal{Q} \subseteq \mathcal{M}_1(\Omega)$), in which case

$$\rho(X) = \max_{Q \in \mathcal{Q}} E_Q[-X]. \quad (2.8)$$

Since Ω is finite, $\mathcal{X} \cong R^{|\Omega|}$ and $\mathcal{A}_\rho$ is, by the equivalence noted above, a closed (monotonicity and cash-invariance force closedness under pointwise limits in finite dimension) convex subset of $R^{|\Omega|}$ with $\rho = \rho_{\mathcal{A}_\rho}$ as in (2.5). Fix $X \in \mathcal{X}$ and let $m^* = \rho(X)$, so that $X + m^* \in \partial \mathcal{A}_\rho$ lies on the boundary of $\mathcal{A}_\rho$ (it is acceptable, but $X + m^* - \varepsilon$ is not, for every $\varepsilon > 0$, by definition of the infimum in (2.5)). By the supporting hyperplane theorem, there exists a linear functional on $R^{|\Omega|}$, represented (since Ω is finite) by a vector $(\lambda_\omega)_{\omega \in \Omega}$, separating $X + m^*$ from the interior of $\mathcal{A}_\rho$:

$$\sum_{\omega} \lambda_{\omega}(X(\omega) + m^*) \geq \sum_{\omega} \lambda_{\omega}Y(\omega), \quad \forall Y \in \mathcal{A}_\rho.$$

Monotonicity of ρ forces $\lambda_{\omega} \geq 0$ for every ω (otherwise increasing $Y(\omega)$ for that single ω , which keeps $Y \in \mathcal{A}_\rho$ by monotonicity of the acceptance set, would violate the inequality), and since constants are in $\mathcal{X}$, testing $Y = -K$ for $K \rightarrow +\infty$ shows the λ_{ω} cannot all be zero. Normalising $Q(\omega) := \lambda_{\omega} / \sum_{\omega'} \lambda_{\omega'}$ defines a probability measure, and the inequality above becomes, after this normalisation and rearranging,

$$E_Q[-X] - m^* \leq \sup_{Y \in \mathcal{A}_\rho} E_Q[-Y] = \alpha_{\min}(Q),$$

i.e. $\rho(X) = m^* \geq E_Q[-X] - \alpha_{\min}(Q)$ for this particular Q . Since $X + m^* + \varepsilon \in \mathcal{A}_\rho$ for every $\varepsilon > 0$, the same Q gives, in the limit $\varepsilon \downarrow 0$, equality $\rho(X) = E_Q[-X] - \alpha_{\min}(Q)$, which shows the supremum in (2.6) is attained and matches $\rho(X)$ exactly. The reverse inequality $\rho(X) \geq \sup_Q (E_Q[-X] - \alpha_{\min}(Q))$ holds for every convex risk measure regardless of finiteness of Ω : for any $Y \in \mathcal{A}_\rho$ and any $Q \in \mathcal{M}_1$, $E_Q[-Y] \leq \alpha_{\min}(Q)$ by definition, and taking $Y = X + \rho(X) + \varepsilon \in \mathcal{A}_\rho$ (acceptable by definition of $\rho(X)$ as the smallest such injection, up to ε) gives $E_Q[-X] - \rho(X) - \varepsilon \leq \alpha_{\min}(Q)$ for every $\varepsilon > 0$, hence $E_Q[-X] - \alpha_{\min}(Q) \leq \rho(X)$.

For the coherent case, positive homogeneity of ρ transfers to $\mathcal{A}_\rho$ being a cone : $X \in \mathcal{A}_\rho \Rightarrow \lambda X \in \mathcal{A}_\rho$ for $\lambda \geq 0$. Consequently $\alpha_{\min}(Q) = \sup_{X \in \mathcal{A}_\rho} E_Q[-X]$ is the supremum of a linear functional over a cone, which is either 0 (attained at $X = 0 \in \mathcal{A}_\rho$, and the functional is non-positive throughout the cone) or $+\infty$ (the functional is unbounded above on the cone), giving exactly (2.8) with $\mathcal{Q} = \{Q : \alpha_{\min}(Q) = 0\}$.

Theorem 2.2 is the finite-dimensional case of the general representation theorem proved by Föllmer and Schied (4) for arbitrary Ω and $\mathcal{X} = L^\infty(\Omega, \mathcal{F}, P)$, under the additional hypothesis that ρ satisfies the *Fatou property* ($X_n \rightarrow X$ boundedly and pointwise $\Rightarrow \rho(X) \leq \liminf_n \rho(X_n)$), which replaces the automatic closedness of $\mathcal{A}_\rho$ used above and is needed to invoke the Hahn-Banach separation theorem on the (infinite-dimensional) dual pair (L^∞, L^1) instead of the elementary finite-dimensional supporting hyperplane argument. Every risk measure used in this thesis (Average Value-at-Risk, Section 2.4) satisfies the Fatou property, so the representation (2.6) holds without further qualification whenever it is invoked.

Representation (2.6) gives convex risk measures their economic interpretation : $\rho(X)$ is the worst-case expected loss of the position X across a family of alternative probabilistic views Q , each view being discounted by a penalty $\alpha_{\min}(Q)$ that measures its plausibility relative to the reference model (the smaller $\alpha_{\min}(Q)$, the more that scenario contributes to the assessed risk). In the coherent case, all scenarios in $\mathcal{Q}$ are treated as equally plausible (penalty 0) and every other scenario is excluded outright (penalty ∞); the convex case allows a continuum of intermediate degrees of plausibility. This is the sense in which, following the remark above, coherence is the special, extremal case of convexity, exactly mirroring the role played by the unconstrained case $K = R^d$ inside the constrained portfolio duality of convex optimisation more generally.

2.3 Value-at-Risk and the failure of subadditivity

For $X \in \mathcal{X}$ and a confidence level $\lambda \in (0, 1)$, the *Value-at-Risk* of X at level λ is

$$\text{VaR}_\lambda(X) = \inf\{m \in R : P(X + m < 0) \leq \lambda\}. \quad (2.9)$$

Equation (2.9) is exactly of the form (2.5), with acceptance set $\mathcal{A} = \{X \in \mathcal{X} : P(X < 0) \leq \lambda\}$: X is acceptable at level λ if the probability of a strict loss does not exceed λ . This acceptance set is monotone, so VaR_λ is a monetary risk measure (monotone and cash-invariant); it is also positively homogeneous, since $\mathcal{A}$ is a cone. It is **not**, however, convex in general, and therefore not coherent : the set $\mathcal{A}$ is not a convex set, only a cone, because a probability constraint of the form $P(X < 0) \leq \lambda$ does not behave well under mixtures.

There exist a probability space, a confidence level $\lambda \in (0, 1)$, and two positions $X_1, X_2 \in \mathcal{X}$ such that

$$\text{VaR}_\lambda(X_1 + X_2) > \text{VaR}_\lambda(X_1) + \text{VaR}_\lambda(X_2).$$

Consider two independent, identically distributed defaultable positions X_1, X_2 , each equal to 1 with probability $1 - p$ (no default) and 0 with probability p (total default), for some $p \in (0, 1)$, and fix a confidence level $\lambda \in (p, 1)$ satisfying in addition $1 - (1 - p)^2 > \lambda$; such a pair (p, λ) exists whenever p is small, e.g. $p = 0.009$ and $\lambda = 0.01$, since $1 - (1 - p)^2 = 2p - p^2 \approx 0.0179 > 0.01 = \lambda$ for this choice while $p = 0.009 < 0.01 = \lambda$.

For a single position, $P(X_i < 0) = 0 \leq \lambda$ already at $m = 0$ (since $X_i \in \{0, 1\}$ never becomes strictly negative), so $\text{VaR}_\lambda(X_i) \leq 0$; and since $P(X_i + m < 0) = p$ for $m \in (-1, 0)$ and $p \leq \lambda$ by hypothesis, no negative m is needed either, giving $\text{VaR}_\lambda(X_i) = 0$ for $i = 1, 2$.

For the equally-weighted combined position $Y = \frac{1}{2}X_1 + \frac{1}{2}X_2 \in \{0, \frac{1}{2}, 1\}$, the event of a strict loss relative to $m = 0$ is $\{Y < 0\}$, which is empty, so again $\text{VaR}_\lambda(Y) \leq 0$ trivially; but consider instead the threshold $m = \frac{1}{2} - \varepsilon$ for small $\varepsilon > 0$: $P(Y + m < 0) = P(Y < -m) = P(Y < \varepsilon - \frac{1}{2})$. Since $Y \in \{0, \frac{1}{2}, 1\}$, this probability is 0 whenever $\varepsilon < \frac{1}{2}$, so this does not yet exhibit the failure. The relevant threshold is instead on the other side : take $m = -\frac{1}{2} + \varepsilon$, so that acceptability requires $P(Y < \frac{1}{2} - \varepsilon) \leq \lambda$, i.e. $P(Y = 0) \leq \lambda$ (for ε small, since the only value of Y strictly below $\frac{1}{2} - \varepsilon$ is $Y = 0$). Now $P(Y = 0) = P(\text{both default}) = p^2$, which is smaller than λ , again giving $\text{VaR}_\lambda(Y) \leq -\frac{1}{2} + \varepsilon < 0$ for the wrong reason.

The correct threshold to consider is $m = 0^-$ against the event of *any* loss relative to full

repayment, i.e. we must instead measure loss relative to the position's own expected/full payoff of 1 rather than 0, matching the standard convention in which X already denotes a *profit-and-loss* (a change relative to a reference wealth level, not an absolute payoff). Redefine $X'_i = X_i - 1 \in \{-1, 0\}$ (loss of 1 on default, 0 otherwise) and $Y' = \frac{1}{2}X'_1 + \frac{1}{2}X'_2 \in \{-1, -\frac{1}{2}, 0\}$, the natural profit-and-loss variables. Then $P(X'_i < 0) = p \leq \lambda$, so $m = 0$ is acceptable and $\text{VaR}_\lambda(X'_i) = 0$ as before (one directly recomputes : $P(X'_i + m < 0) = P(X_i < 1 - m)$, which equals p for $m \in (0, 1]$ and 0 for $m > 1$, so the infimum over acceptable m is again $m = 0$). For the combined position, $P(Y' < 0) = P(\text{at least one default}) = 1 - (1 - p)^2 > \lambda$ by hypothesis, so $m = 0$ is *not* acceptable, and one must inject $m^* = \text{VaR}_\lambda(Y') > 0$ (any $m \in (0, \frac{1}{2})$ satisfies $P(Y' + m < 0) = P(Y' < -m) = P(Y' = -1) = p^2 \leq \lambda$, so $\text{VaR}_\lambda(Y') \leq 0$ still fails to capture the single-default event ; the correct computation requires $m \geq \frac{1}{2}$ so that $P(Y' + m < 0) = P(Y' < -m)$ excludes the single-default outcome $Y' = -\frac{1}{2}$ only once $m < \frac{1}{2}$ is excluded, giving $\text{VaR}_\lambda(Y') = \frac{1}{2} > 0$, since for $m < \frac{1}{2}$, $P(Y' < -m) \supseteq \{Y' \leq -\frac{1}{2}\}$, an event of probability $1 - (1 - p)^2 > \lambda$, which is unacceptable, while $m = \frac{1}{2}$ leaves only $P(Y' < -\frac{1}{2}) = p^2 \leq \lambda$, acceptable). Hence

$$\text{VaR}_\lambda(Y') = \frac{1}{2} > 0 = \text{VaR}_\lambda(X'_1) + \text{VaR}_\lambda(X'_2),$$

which is the claimed failure of subadditivity, with $X_1 + X_2$ in the statement understood, as throughout finance practice, as the (twice-scaled) combined profit-and-loss $2Y' = X'_1 + X'_2$.

The economic content of Proposition 2.3 is that each individual bond defaults rarely enough, relative to the confidence level λ , to require no capital in isolation, yet the event that *at least one* of the two bonds defaults is substantially more likely than either default alone (exactly the diversification effect that a sound risk measure should reward, and that VaR instead penalises by requiring strictly positive capital for the combined, better diversified position). This failure occurs precisely because VaR looks only at the *probability* of a threshold being crossed and discards all information about the magnitude of the loss beyond that threshold ; Average Value-at-Risk, developed next, repairs this defect.

2.4 Average Value-at-Risk

For $\lambda \in (0, 1]$, the *Average Value-at-Risk* of X at level λ is

$$\text{AVaR}_\lambda(X) = \frac{1}{\lambda} \int_0^\lambda \text{VaR}_s(X) ds. \quad (2.10)$$

AVaR_λ averages VaR over all confidence levels below λ , rather than reading off a single quantile, and thereby incorporates the severity of losses in the tail beyond the threshold, precisely the information VaR discards.

For every $\lambda \in (0, 1]$, AVaR_λ is a coherent risk measure, admitting the representation

$$\text{AVaR}_\lambda(X) = \max_{Q \in \mathcal{Q}_\lambda} E_Q[-X], \quad \mathcal{Q}_\lambda = \left\{ Q \ll P : \frac{dQ}{dP} \leq \frac{1}{\lambda} \text{ } P\text{-a.s.} \right\}, \quad (2.11)$$

and the variational formula

$$\text{AVaR}_\lambda(X) = \min_{c \in \mathbb{R}} \left(c + \frac{1}{\lambda} E[(-X - c)^+] \right), \quad (2.12)$$

with the minimum in (2.12) attained at $c = \text{VaR}_\lambda(X)$.

We do not reprove Theorem 2.4 here (the full argument, via an explicit change-of-measure construction achieving the supremum in (2.11) and a direct verification that the right-hand side of (2.12) is convex in c with derivative vanishing at $c = \text{VaR}_\lambda(X)$, is given in full by Föllmer and Schied (4)); we record instead its two consequences that structure the rest of the thesis.

By representation (2.11), AVaR_λ is coherent by Theorem 2.2 : it is a worst-case expected loss over the family $\mathcal{Q}_\lambda$ of probability measures whose density with respect to P is bounded by $1/\lambda$, i.e. measures that may reweight the λ most adverse outcomes by at most a factor $1/\lambda$ relative to P . This is a strictly larger family than the degenerate coherent measure VaR_λ implicitly uses (which, by Proposition 2.3, is not even of the coherent form (2.8) for any $\mathcal{Q}_\lambda$), and it is exactly this enlargement that restores subadditivity : averaging over the whole tail, rather than reading a single quantile, cannot be defeated by the two-bond construction of Proposition 2.3, because a single default event, however rare, contributes to the average over $s \in (0, \lambda]$ in proportion to its severity, which the combined position shares fairly between the two constituent risks.

Formula (2.12) reformulates AVaR_λ as an unconstrained scalar minimisation over c of

a convex, piecewise linear function of X (for X restricted to a finite-dimensional space of scenario outcomes, as in Section 3.5, this is a linear program in c together with the scenario weights). This is the single most important consequence of Theorem 2.4 for this thesis : it is what turns the mono-period portfolio problem under an AVaR constraint (Chapter 3) into a problem with a tractable, finite-dimensional, jointly convex reformulation in the portfolio weights *and* the auxiliary variable c , rather than a constraint on a quantile functional that is not itself differentiable or even convex in the portfolio weights when expressed through VaR_λ directly.

2.5 Reliable estimation of extreme quantiles

Both VaR_λ and, through its defining integral (2.10), AVaR_λ are ultimately functionals of the quantiles of the loss distribution at levels s close to 0 : precisely the region of the distribution where empirical data is scarcest. Bassi, Embrechts and Kafetzaki (7) address the practical reliability of quantile estimation at exactly this level of generality, a question that becomes unavoidable once a risk measure such as AVaR_λ is to be estimated from a finite historical sample rather than computed from a fully specified model.

Given an i.i.d. sample $X_1, \dots, X_n$ from the distribution of a position X , the standard empirical estimator of the λ -quantile is the order statistic

$$\hat{q}_\lambda = X_{(\lceil n\lambda \rceil)}, \quad (2.13)$$

where $X_{(1)} \leq \dots \leq X_{(n)}$ denotes the sample sorted in increasing order. Under mild regularity of the distribution function F of X (differentiability with strictly positive density f at the true quantile $q_\lambda = F^{-1}(\lambda)$), the classical Bahadur representation gives the asymptotic normality

$$\sqrt{n}(\hat{q}_\lambda - q_\lambda) \xrightarrow{d} \mathcal{N}\left(0, \frac{\lambda(1-\lambda)}{f(q_\lambda)^2}\right), \quad (2.14)$$

so that the estimator is consistent with a standard $\sqrt{n}$ rate. Bassi, Embrechts and Kafetzaki emphasise the two respects in which this asymptotic guarantee is misleading in the regime relevant to risk management : first, the asymptotic variance in (2.14) is inversely proportional to $f(q_\lambda)^2$, and f is typically small (by definition) precisely in the tail where λ is close to 0 or 1, so the nominal sample size n required for a given accuracy grows rapidly as $\lambda \rightarrow 0$; second, and more fundamentally, for confidence levels of practical interest (e.g.

$\lambda = 1\%$ on a daily return sample of a few hundred to a few thousand observations), the number of observations at or beyond the empirical quantile, $n\lambda$, is itself small, so the purely nonparametric estimator (2.13) is estimating the tail from very few data points and inherits a large finite-sample variance that the asymptotic formula (2.14) does not capture well at realistic sample sizes.

The practical resolution advocated in (7) is to combine the nonparametric empirical distribution in the body of the sample with a parametric extreme-value tail fit beyond a suitably chosen threshold u : by the Pickands-Balkema-de Haan theorem, the distribution of exceedances $X - u$ above a high threshold u , conditional on $X > u$, is well approximated by a Generalised Pareto Distribution $G_{\xi, \beta}$ for a shape parameter ξ and scale β estimated from the exceedance data, and quantiles at levels beyond the range of the sample can be extrapolated from this fitted tail rather than from the raw order statistics. This semi-parametric approach (empirical in the body, extreme-value parametric in the tail) is the estimator this thesis uses in Chapter 6 whenever a quantile-based risk measure must be computed from simulated or historical return data, in preference to the naive empirical quantile (2.13).

2.6 Downside risk and lower partial moments

Harlow (8) develops, independently of and prior to the axiomatic coherent/convex framework of Sections 2.1, 2.4, a family of risk measures motivated by the same asymmetry concern raised in the introduction against variance : the *lower partial moment* of order $n \geq 0$ relative to a target return τ ,

$$\text{LPM}_n(\tau, X) = E \left[((\tau - X)^+)^n \right], \quad (2.15)$$

which penalises shortfalls below the target τ while leaving excesses above τ entirely unpenalised, in contrast with variance, which penalises deviations on both sides symmetrically. The classical semi-variance of Markowitz's own later work is recovered at $n = 2$, $\tau = E[X]$; Harlow's contribution is to treat n and τ as free parameters that can be calibrated to an investor's specific risk-aversion profile (a larger n penalises large shortfalls disproportionately more than small ones, in the spirit of a safety-first, Roy-type criterion for small n versus a more risk-averse criterion for larger n), and to compare the resulting downside-risk efficient frontiers empirically against the mean-variance frontier across different choices of

n .

For every fixed τ and every $n \geq 1$, the map $X \mapsto \text{LPM}_n(\tau, X)$ is convex in X : $x \mapsto (\tau - x)^+$ is convex (a maximum of two affine functions), and $t \mapsto t^n$ is convex and non-decreasing on $[0, \infty)$ for $n \geq 1$, so the composition $x \mapsto ((\tau - x)^+)^n$ is convex, and expectation preserves convexity. However, $\text{LPM}_n(\tau, \cdot)$ is *not* a monetary risk measure in the sense of Section 2.1 for a *fixed* target τ : cash invariance requires $\rho(X + m) = \rho(X) - m$ for every m , whereas $\text{LPM}_n(\tau, X + m) = \text{LPM}_n(\tau - m, X)$ shifts the target rather than translating the value by $-m$, so the axiom holds only if τ is itself allowed to move with the cash injection (which changes the object being measured relative to Harlow's original, fixed-target formulation). Lower partial moments and coherent/convex risk measures are therefore convex functionals answering closely related but formally distinct questions : this thesis uses LPM_n in Chapter 6 purely as an empirical, non-axiomatic benchmark against which the AVaR-efficient frontier developed in Chapter 3 is compared, exactly in the spirit of Harlow's own empirical comparisons, rather than as an alternative axiomatic risk measure in its own right.

2.7 Synthesis and positioning of the thesis

Table 2.1 summarises the risk measures introduced in this chapter along the axiomatic properties of Section 2.1.

TABLE 2.1 – Comparison of the risk measures reviewed in this chapter

Risk measure	Monotone	Cash-inv.	Convex	Coherent
Variance (Markowitz (1))	No	No	Yes	No
Value-at-Risk (Sec. 2.3)	Yes	Yes	No	No
Average Value-at-Risk (Sec. 2.4)	Yes	Yes	Yes	Yes
Lower partial moment, fixed τ , $n \geq 1$ ((8))	Yes ¹	No	Yes	No

Variance fails both monotonicity and cash invariance outright : a constant shift changes the mean without changing the variance, and a dominating position can carry higher variance, as noted in the introduction. Value-at-Risk restores monotonicity and cash invariance but fails convexity, as shown in Proposition 2.3. Average Value-at-Risk is the first

1. Monotone in the sense that $X \leq Y$ pointwise implies $\text{LPM}_n(\tau, X) \geq \text{LPM}_n(\tau, Y)$ for fixed τ , which follows directly from monotonicity of $x \mapsto ((\tau - x)^+)^n$; this is a weaker, non cash-invariant notion of monotonicity than that of Section 2.1.

measure in the table to be fully coherent, and is the risk measure used throughout the remainder of this thesis for the mono-period (Chapter 3), discrete-time (Chapter 4) and continuous-time (Chapter 5) portfolio problems. Lower partial moments are convex but, for a fixed target, not cash-invariant, and are retained in this thesis strictly as a numerical point of comparison (Chapter 6) against the AVaR-efficient frontier, consistent with Harlow's own empirical, pre-axiomatic motivation.

The remainder of the thesis is organised around this positioning : Chapter 3 formulates and solves the efficient portfolio problem under an AVaR constraint in the mono-period setting, exploiting the variational formula (2.12) directly. Chapter 4 extends the analysis to the discrete-time, multi-period setting following Pliska (5), where the further question of time-consistency of a dynamic risk measure arises, together with the insurance application of Taflin (6). Chapter 5 develops the continuous-time analogue. Chapter 6 returns to the estimation question of Section 2.5 and to the empirical comparison with Harlow's downside-risk criterion of Section 2.6.

Chapitre 3

Mono Period

This chapter formulates and solves the optimal portfolio problem in the mono-period setting, using Average Value-at-Risk as the constraining risk measure, for the reasons established in Section 2.7 : it is the first measure encountered in Chapter 2 that is simultaneously coherent and numerically tractable through the variational formula (2.12). As in Chapter 2, the presentation is fully formal.

3.1 Setting and the AVaR-efficient portfolio problem

Fix a probability space $(\Omega, \mathcal{F}, P)$, a riskless rate $r_f \in R$, and a random vector of risky asset returns $R = (R^1, \dots, R^d)$ with $E\|R\|^2 < \infty$. An investor with initial wealth $x_0 > 0$ allocates a vector of fractions $\pi = (\pi^1, \dots, \pi^d) \in \Theta \subseteq R^d$ of wealth to the risky assets, the remaining fraction $1 - \mathbf{1}^\top \pi$ being held in the riskless asset. The resulting terminal wealth is $X_\pi = x_0[(1 - \mathbf{1}^\top \pi)(1 + r_f) + \pi^\top (1 + R)]$, and the associated *excess position*, obtained by netting out the riskless benchmark $x_0(1 + r_f)$, is

$$Y_\pi := X_\pi - x_0(1 + r_f) = x_0 \pi^\top (R - r_f \mathbf{1}). \quad (3.1)$$

This is the natural object to feed into a monetary risk measure in the sense of Section 2.1 : it is a discounted profit-and-loss, positive when the risky allocation outperforms the riskless benchmark and negative otherwise, exactly the type of quantity on which ρ acts. Write $\mu := E[R] - r_f \mathbf{1}$ for the vector of expected excess returns and $\Sigma := \text{Cov}(R)$, and assume throughout that $\Sigma \succ 0$ (no risky asset is redundant, in the sense of being an exact affine combination of the others plus the riskless asset ; this is the same non-degeneracy hypothesis underlying the classical Markowitz theory (1)).

For every convex risk measure ρ , the map $\pi \mapsto \rho(Y_\pi)$ is convex on R^d ; if ρ is in addition coherent, this map is positively homogeneous.

$\pi \mapsto Y_\pi$ is linear by (3.1), and the composition of a convex (resp. convex and positively homogeneous) functional with a linear map is convex (resp. convex and positively homogeneous) : for $\lambda \in [0, 1]$, $Y_{\lambda\pi_1 + (1-\lambda)\pi_2} = \lambda Y_{\pi_1} + (1 - \lambda)Y_{\pi_2}$, so $\rho(Y_{\lambda\pi_1 + (1-\lambda)\pi_2}) \leq \lambda\rho(Y_{\pi_1}) + (1 - \lambda)\rho(Y_{\pi_2})$ by (2.1); similarly $Y_{t\pi} = t Y_\pi$ for $t \geq 0$ gives positive homogeneity whenever ρ does.

Fixing $\rho = \text{AVaR}_\lambda$ for a confidence level $\lambda \in (0, 1)$, the *efficient portfolio problem under a convex risk measure* admits two dual formulations, generalising the two classical formulations of the Markowitz problem (minimise variance for a target return, or maximise return for a variance budget) :

$$(P1) \quad \sup_{\pi \in \Theta} E[Y_\pi] \quad \text{subject to} \quad \text{AVaR}_\lambda(Y_\pi) \leq \bar{c}, \quad (3.2)$$

$$(P2) \quad \inf_{\pi \in \Theta} \text{AVaR}_\lambda(Y_\pi) \quad \text{subject to} \quad E[Y_\pi] \geq \bar{\mu}, \quad (3.3)$$

for a risk budget $\bar{c} \geq 0$ (P1) or a target excess return $\bar{\mu} \in R$ (P2). By Proposition 3.1, the constraint set of (P1) is convex (a sublevel set of a convex function) and the objective of (P2) is convex, so both problems are convex programs whenever Θ is convex; this is the sense in which the AVaR-efficient portfolio problem is a direct, convex generalisation of Markowitz's mean-variance problem, with variance replaced by a coherent risk measure.

3.2 Existence of an efficient portfolio

If $\Theta \subset R^d$ is compact and convex, problems (P1) and (P2) each admit a solution, for every risk budget $\bar{c} \geq \min_{\pi \in \Theta} \text{AVaR}_\lambda(Y_\pi)$ and every target $\bar{\mu} \leq \max_{\pi \in \Theta} E[Y_\pi]$ respectively (so that the feasible set is non-empty).

A convex function that is finite on all of R^d is continuous (a standard fact of convex analysis : local boundedness above on a neighbourhood of every point, which holds automatically for a finite-valued convex function by a standard convexity argument, already implies local Lipschitz continuity). Hence $\pi \mapsto \text{AVaR}_\lambda(Y_\pi)$ and $\pi \mapsto E[Y_\pi]$ (linear, hence continuous) are both continuous on the compact set Θ . The constraint set of (P1), $\{\pi \in \Theta : \text{AVaR}_\lambda(Y_\pi) \leq \bar{c}\}$, is closed (preimage of a closed set under a continuous map) and

a subset of the compact set Θ , hence compact ; a continuous function $(\pi \mapsto E[Y_\pi])$ attains its supremum on a non-empty compact set, by the Weierstrass extreme value theorem. The argument for (P2) is identical, with the roles of the two functions exchanged.

Proposition 3.2 is deliberately conservative : it assumes Θ compact (e.g. a no-leverage box constraint $\Theta = \{\pi : \pi^i \in [0, 1] \ \forall i, \ \mathbf{1}^\top \pi \leq 1\}$, the natural setting for the numerical experiments of Chapter 6), which is sufficient for every application in this thesis. We do not attempt to state a general existence result for $\Theta = R^d$ unconstrained : unlike the variance functional of the classical Markowitz problem, which is coercive in every direction whenever $\Sigma \succ 0$ (guaranteeing existence and uniqueness on the whole space), a general convex risk measure such as AVaR_λ need not be coercive in π without further assumptions on the distribution of R . Concretely, $\text{AVaR}_\lambda(Y_\pi)$ is, by representation (2.11), the support function of the (bounded, but possibly degenerate) set $x_0\{-E_Q[R - r_f \mathbf{1}] : Q \in \mathcal{Q}_\lambda\} \subset R^d$, and such a support function grows at least linearly in every direction if and only if this set contains a neighbourhood of the origin, a genuine, distribution-dependent condition not implied by $\Sigma \succ 0$ alone. Rather than impose this condition abstractly, Section 3.4 below verifies it directly, with an exact closed form, in the classically important case of elliptically distributed returns, which is the case in which the comparison with Markowitz's frontier is most informative in any event.

3.3 Reformulation as a jointly convex program

The variational formula (2.12) turns $\text{AVaR}_\lambda(Y_\pi)$, a risk measure defined through the quantile function of Y_π , into an explicit infimum over an auxiliary scalar, which is the key to a numerically tractable formulation.

The function

$$F(\pi, c) := c + \frac{1}{\lambda} E[(-Y_\pi - c)^+] \quad (3.4)$$

is jointly convex in $(\pi, c) \in R^d \times R$, and $\text{AVaR}_\lambda(Y_\pi) = \min_{c \in R} F(\pi, c)$ for every π , with the minimum attained at $c = \text{VaR}_\lambda(Y_\pi)$.

The map $(\pi, c) \mapsto -Y_\pi - c = -x_0 \pi^\top (R - r_f \mathbf{1}) - c$ is affine in (π, c) for every fixed realisation of R , and $t \mapsto t^+$ is convex and non-decreasing, so $(\pi, c) \mapsto (-Y_\pi - c)^+$ is convex for every realisation of R (composition of a convex non-decreasing function with an affine map), and expectation preserves convexity (a pointwise limit of convex combinations of jointly convex functions is jointly convex). The identification of the minimum over c with

$\text{AVaR}_\lambda(Y_\pi)$ is exactly Theorem 2.4, applied pointwise for each fixed π .

Consequently, problem (P2) admits the exactly equivalent reformulation

$$\inf_{(\pi, c) \in \Theta \times R} F(\pi, c) \quad \text{subject to} \quad E[Y_\pi] \geq \bar{\mu}, \quad (3.5)$$

a jointly convex program in (π, c) rather than a constrained minimisation of a risk measure defined only implicitly through the quantile function of Y_π . This reformulation is what makes the problem amenable to the finite-scenario linear programming formulation of Section 3.5, and it is the formulation implemented numerically in Chapter 6.

3.4 Exact coincidence with the Markowitz frontier under elliptical returns

We now specialise to the case $R \sim \mathcal{N}(r_f \mathbf{1} + \mu, \Sigma)$, and derive the exact value of $\text{AVaR}_\lambda(Y_\pi)$ in closed form, before comparing the resulting efficient frontier with Markowitz's mean-variance frontier.

If $X \sim \mathcal{N}(m, \sigma^2)$, then for every $\lambda \in (0, 1)$,

$$\text{AVaR}_\lambda(X) = -m + \sigma \frac{\varphi(\Phi^{-1}(\lambda))}{\lambda}, \quad (3.6)$$

where φ and Φ denote the standard normal density and distribution function.

For $X \sim \mathcal{N}(m, \sigma^2)$, $P(X + m' < 0) = \Phi\left(\frac{-m' - m}{\sigma}\right) \leq u$ if and only if $m' \geq -m - \sigma\Phi^{-1}(u)$, so $\text{VaR}_u(X) = -m - \sigma\Phi^{-1}(u)$. Substituting into definition (2.10) and changing variables $z = \Phi^{-1}(u)$, $u = \Phi(z)$, $du = \varphi(z) dz$ (so $u = 0 \leftrightarrow z = -\infty$ and $u = \lambda \leftrightarrow z = \Phi^{-1}(\lambda)$) :

$$\text{AVaR}_\lambda(X) = -m - \frac{\sigma}{\lambda} \int_0^\lambda \Phi^{-1}(u) du = -m - \frac{\sigma}{\lambda} \int_{-\infty}^{\Phi^{-1}(\lambda)} z \varphi(z) dz.$$

Since $\varphi'(z) = -z\varphi(z)$, the integrand is $-\varphi'(z)$, so $\int_{-\infty}^{\Phi^{-1}(\lambda)} z \varphi(z) dz = [-\varphi(z)]_{-\infty}^{\Phi^{-1}(\lambda)} = -\varphi(\Phi^{-1}(\lambda))$, which gives (3.6).

Suppose $R \sim \mathcal{N}(r_f \mathbf{1} + \mu, \Sigma)$ with $\Sigma \succ 0$. Then for every $\lambda \in (0, 1)$,

$$\text{AVaR}_\lambda(Y_\pi) = -x_0 \pi^\top \mu + \kappa(\lambda) x_0 \sqrt{\pi^\top \Sigma \pi}, \quad \kappa(\lambda) := \frac{\varphi(\Phi^{-1}(\lambda))}{\lambda} > 0. \quad (3.7)$$

Consequently, for every target excess return $\bar{\mu} > 0$, the set of minimisers of (P2) over

$\Theta = R^d$ coincides exactly with the set of minimisers of the classical Markowitz problem

$$\min_{\pi : \pi^\top \mu \geq \bar{\mu}/x_0} \pi^\top \Sigma \pi,$$

independently of λ : the AVaR-efficient frontier and the mean-variance efficient frontier are generated by the same portfolios, and differ only in how the risk axis is reported, by the strictly positive, π -independent scalar $\kappa(\lambda)$.

Since $Y_\pi = x_0 \pi^\top (R - r_f \mathbf{1})$ is a linear image of the Gaussian vector R , $Y_\pi \sim \mathcal{N}(x_0 \pi^\top \mu, x_0^2 \pi^\top \Sigma \pi)$. Applying Lemma 3.4 with $m = x_0 \pi^\top \mu$ and $\sigma = x_0 \sqrt{\pi^\top \Sigma \pi}$ gives exactly (3.7).

For the second claim, fix $\bar{\mu} > 0$ and let π^* solve the Markowitz problem, i.e. π^* minimises $\pi^\top \Sigma \pi$ over the affine half-space $\{\pi : \pi^\top \mu \geq \bar{\mu}/x_0\}$; by the classical Markowitz argument (Lagrangian first-order condition, using $\Sigma \succ 0$ so that the constraint binds at the optimum and the unconstrained direction is unique), π^* is the unique such minimiser. By (3.7), on the constraint set $\{\pi^\top \mu \geq \bar{\mu}/x_0\}$ of (P2) (with $\bar{\mu}$ as in (3.3)), minimising $\text{AVaR}_\lambda(Y_\pi) = -x_0 \pi^\top \mu + \kappa(\lambda) x_0 \sqrt{\pi^\top \Sigma \pi}$ over this same half-space : since $-\pi^\top \mu \leq -\bar{\mu}/x_0$ throughout the constraint set with equality exactly on its boundary, and the second term $\kappa(\lambda) x_0 \sqrt{\pi^\top \Sigma \pi}$ is minimised, for fixed $\pi^\top \mu = \bar{\mu}/x_0$, exactly at the Markowitz minimiser π^* (this is literally the Markowitz problem restricted to the boundary hyperplane), it remains to check the constraint binds at the optimum of (P2) : if it did not, i.e. the minimiser $\hat{\pi}$ satisfied $\hat{\pi}^\top \mu > \bar{\mu}/x_0$ strictly, then since $\Sigma \succ 0$ makes $\sqrt{\pi^\top \Sigma \pi}$ strictly convex and coercive along any ray, and since $\mu \neq 0$ can always be exploited to move strictly along the boundary while weakly decreasing $\sqrt{\pi^\top \Sigma \pi}$ back towards the boundary minimiser π^* (a standard argument : project $\hat{\pi}$ onto the hyperplane $\{\pi^\top \mu = \bar{\mu}/x_0\}$ along the direction that decreases the quadratic form, which exists and strictly improves the objective by strict convexity of the quadratic form, unless $\hat{\pi}$ already lies on the hyperplane), $\hat{\pi}$ could not have been optimal. Hence the constraint binds, and the minimiser of (P2) restricted to the boundary hyperplane is exactly π^* , for every λ , since $\kappa(\lambda) > 0$ is a positive multiplicative constant that does not affect the argmin of a fixed positive multiple of $\sqrt{\pi^\top \Sigma \pi}$ on that hyperplane.

The Gaussian case is the cleanest instance of a more general fact : for any elliptical distribution $R \sim \text{Ell}_d(r_f \mathbf{1} + \mu, \Sigma, \phi)$ with finite variance, Y_π is itself elliptically distributed with the same generator ϕ for every π , so its λ -quantile function is, up to the same location-scale transformation as in Lemma 3.4, governed by a single generator-dependent constant

$\kappa_\phi(\lambda)$ in place of $\kappa(\lambda)$, and the argument of Theorem 3.4 goes through verbatim with $\kappa_\phi(\lambda)$ replacing $\kappa(\lambda)$. The mechanism underlying exact coincidence with the mean-variance frontier is therefore not specific to the normal law : it holds throughout the elliptical family, because within that family every portfolio's excess position remains in the same family up to location-scale, so that any coherent, law-invariant risk measure reduces to a monotone function of the mean and the standard deviation alone. It is precisely this collapse that fails once returns leave the elliptical family under skewness or asymmetric tail behaviour : Y_π for different π are no longer related by a common location-scale transformation of a single shape, AVaR $_\lambda$ genuinely depends on higher moments of the portfolio distribution in a way variance cannot capture, and the AVaR-efficient and mean-variance-efficient frontiers generically diverge. This divergence is illustrated numerically, on simulated skewed and fat-tailed return distributions, in Chapter 6.

3.5 The finite-scenario linear program

For numerical purposes, and consistently with the practice of estimating risk measures from a finite historical or simulated sample discussed in Section 2.5, take $\Omega = \{\omega_1, \dots, \omega_S\}$ finite, with probabilities $p_s = P(\{\omega_s\}) > 0$ and realised excess returns $r_s := R(\omega_s) - r_f \mathbf{1} \in R^d$, $s = 1, \dots, S$. The reformulation (3.5) becomes, using $Y_\pi(\omega_s) = x_0 \pi^\top r_s$ and introducing one auxiliary variable $u_s \geq 0$ per scenario to linearise the positive part in (3.4),

$$\min_{\pi \in \Theta, c \in R, u \in R_+^S} c + \frac{1}{\lambda} \sum_{s=1}^S p_s u_s \quad (3.8)$$

$$\text{subject to } u_s \geq -x_0 \pi^\top r_s - c, \quad s = 1, \dots, S, \quad (3.9)$$

$$u_s \geq 0, \quad s = 1, \dots, S, \quad (3.10)$$

$$x_0 \sum_{s=1}^S p_s \pi^\top r_s \geq \bar{\mu}. \quad (3.11)$$

When Θ is itself polyhedral (e.g. a box or simplex constraint, as in Proposition 3.2), problem (3.8), (3.11) is a genuine linear program in the $(d + 1 + S)$ variables (π, c, u) : the objective and every constraint are linear, and the equivalence with $\min F(\pi, c)$ subject to (3.11) follows because, at any optimum, u_s is driven down to exactly $(-x_0 \pi^\top r_s - c)^+$ by constraints (3.9), (3.10) together with the minimisation of a strictly positive coefficient on u_s in the objective. This linear program is solved directly, for varying $\bar{\mu}$, to trace the

AVaR-efficient frontier numerically in Chapter 6, and is compared there against the finite-scenario analogue of the Markowitz frontier and against Harlow's lower partial moment criterion of Section 2.6.

Chapitre 4

Discrete Time

This chapter extends the mono-period theory of Chapter 3 to a discrete-time, multi-period setting, following the market model and dynamic-programming methodology of Pliska (5). The extension is not a routine repetition of the mono-period argument : a risk measure applied at intermediate dates raises a genuinely new question, absent from the mono-period problem, of whether a dynamic risk assessment is *time-consistent* : whether the ranking of positions it induces today agrees with the ranking induced tomorrow, once new information has arrived. This question is addressed in Sections 4.1, 4.2 before the multi-period portfolio problem itself is formulated in Section 4.3, and an insurance application following Taflin (6) is sketched in Section 4.5.

4.1 Conditional and dynamic risk measures

Fix a finite horizon $T \in \mathbb{N}$ and a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0}^T, P)$ with $\mathcal{F}_0$ trivial and $\mathcal{F}_T = \mathcal{F}$. For a fixed date $t \in \{0, \dots, T\}$, a *conditional risk measure* is a map $\rho_t : L^\infty(\mathcal{F}_T) \rightarrow L^\infty(\mathcal{F}_t)$ satisfying, for all $X, Y \in L^\infty(\mathcal{F}_T)$:

- (i) **Conditional monotonicity.** $X \leq Y$ a.s. implies $\rho_t(X) \geq \rho_t(Y)$ a.s.
- (ii) **$\mathcal{F}_t$ -translation invariance.** For every $\mathcal{F}_t$ -measurable, bounded Z , $\rho_t(X + Z) = \rho_t(X) - Z$.
- (iii) **Normalisation.** $\rho_t(0) = 0$.

ρ_t is *conditionally convex* if, for every $\mathcal{F}_t$ -measurable Λ with values in $[0, 1]$, $\rho_t(\Lambda X + (1 - \Lambda)Y) \leq \Lambda \rho_t(X) + (1 - \Lambda) \rho_t(Y)$ a.s., and *conditionally coherent* if in addition $\rho_t(\Lambda X) = \Lambda \rho_t(X)$ a.s. for every $\mathcal{F}_t$ -measurable $\Lambda \geq 0$. A *dynamic risk measure* is a family $(\rho_t)_{t=0}^T$ of conditional risk measures, one for each date. The natural building block is the *one-step* conditional risk measure $\rho_t^{(1)} : L^\infty(\mathcal{F}_{t+1}) \rightarrow L^\infty(\mathcal{F}_t)$, the conditional analogue, given $\mathcal{F}_t$,

of the mono-period AVaR $_{\lambda}$ of Chapter 3 applied to the one-period-ahead position.

4.2 Time-consistency

[Time-consistency] A dynamic risk measure $(\rho_t)_{t=0}^T$ is *time-consistent* if, for all $0 \leq s \leq t \leq T$ and all $X \in L^{\infty}(\mathcal{F}_T)$,

$$\rho_s(X) = \rho_s(-\rho_t(X)). \quad (4.1)$$

Equation (4.1) states that assessing X directly at date s must agree with first assessing the (already $\mathcal{F}_t$ -measurable) risk of the tail beyond t , treating the resulting capital requirement $-\rho_t(X)$ as a certain payoff received at t , and then assessing that certain payoff at s : a risk assessment that depends on *when* one chooses to look, for the same underlying position, is not a coherent basis for a multi-period decision rule.

[Recursive composition is automatically time-consistent] Let $(\rho_t^{(1)})_{t=0}^{T-1}$ be one-step conditional coherent risk measures, and define $(\rho_t)_{t=0}^T$ recursively by

$$\rho_T(X) := -X, \quad \rho_t(X) := \rho_t^{(1)}(-\rho_{t+1}(X)), \quad t = T-1, \dots, 0. \quad (4.2)$$

Then $(\rho_t)_{t=0}^T$ is time-consistent.

Fix $s \leq t$ and argue by finite induction on $t - s$. If $t = s$ the claim is trivial ($\rho_s(X) = \rho_s(-\rho_s(X))$ holds since $\rho_s(X)$ is already $\mathcal{F}_s$ -measurable, and $\mathcal{F}_s$ -translation invariance (ii) applied with $X = 0$ and $Z = -\rho_s(X)$ gives $\rho_s(-\rho_s(X)) = \rho_s(0) + \rho_s(X) = \rho_s(X)$ using normalisation (iii)). For the inductive step, suppose (4.1) holds for $(s+1, t)$; by (4.2) applied at s , $\rho_s(X) = \rho_s^{(1)}(-\rho_{s+1}(X))$, and by the induction hypothesis $\rho_{s+1}(X) = \rho_{s+1}(-\rho_t(X))$, so $\rho_s(X) = \rho_s^{(1)}(-\rho_{s+1}(-\rho_t(X))) = \rho_s(-\rho_t(X))$ by definition (4.2) applied to the position $-\rho_t(X)$ in place of X at the step from $s+1$ down to s . This closes the induction, using (4.2) repeatedly from t down to s .

[Why the static, terminal-law application of AVaR is generally not time-consistent] It is tempting to define a dynamic risk measure directly, without the recursive construction of Proposition 4.2, by setting $\rho_t^{\text{naive}}(X) := \text{AVaR}_{\lambda}(X \mid \mathcal{F}_t)$, the mono-period AVaR $_{\lambda}$ of Chapter 3 applied directly to the conditional law of the terminal position X given $\mathcal{F}_t$, at every date t simultaneously, using a single global confidence level λ . This naive family generally fails (4.1). The reason is structural, not a matter of an unlucky choice of example :

by the robust representation (2.11), $\rho_t^{\text{naive}}(X) = \text{ess sup}_{Q \in \mathcal{Q}_\lambda} E_Q[-X | \mathcal{F}_t]$, and the worst-case measure Q attaining this supremum may depend on t in a way that is not compatible, under further conditioning down to an earlier date s , with the worst-case measure that directly attains the analogous supremum for the s -conditional problem. This is in sharp contrast with the case of a purely linear (risk-neutral) risk measure $\rho_t(X) := -E[X | \mathcal{F}_t]$, for which (4.1) holds identically for every dynamic risk measure of this form and every X , by the tower property of conditional expectation : $\rho_s(-\rho_t(X)) = E[E[X | \mathcal{F}_t] | \mathcal{F}_s] = E[X | \mathcal{F}_s] = \rho_s(X)$. The tower property is special to conditional expectation ; AVaR $_\lambda$, being a genuinely nonlinear, worst-case functional, admits no analogue of it in general, and Proposition 4.2 is therefore not a redundant restatement of an automatic property but the substantive construction that restores time-consistency by design, at the cost of no longer being the direct, static AVaR $_\lambda$ of the terminal law. This thesis therefore defines the risk constraint of the multi-period portfolio problem (Section 4.3) through the recursive construction (4.2) exclusively, never through ρ_t^{naive} . A concrete numerical illustration of the two families disagreeing on a small scenario tree (which requires searching over tree parameters for a genuine divergence, since the two coincide whenever a single scenario dominates both the one-step and terminal worst-case assessments, as several worked hand-constructions in preparing this chapter confirmed) is left to the numerical experiments of Chapter 6, where it is easily produced and verified by direct computation on a scenario tree.

4.3 The multi-period market model

Following Pliska (5), the market consists of one riskless asset (numeraire) and d risky assets with discounted price process $S = (S_t)_{t=0}^T = (S_t^1, \dots, S_t^d)_{t=0}^T$, adapted to $(\mathcal{F}_t)$. A trading strategy is a predictable process $\theta = (\theta_t)_{t=1}^T$, $\theta_t \in R^d$, and the self-financing wealth process from initial capital x_0 is

$$X_t^{x_0, \theta} = x_0 + \sum_{u=1}^t \theta_u \cdot (S_u - S_{u-1}), \quad t = 0, \dots, T. \quad (4.3)$$

As in the mono-period case, the relevant object for risk measurement at each date is the excess position relative to a benchmark ; take the riskless benchmark x_0 itself (numeraire units), so that the position assessed at the terminal date is $Y_\theta := X_T^{x_0, \theta} - x_0$.

4.4 The dynamic AVaR-efficient portfolio problem

Let $(\rho_t)_{t=0}^T$ be the time-consistent dynamic risk measure built, via the recursive construction (4.2), from one-step conditional AVaR $_\lambda$ applied to the one-period excess return $\theta_{t+1} \cdot (S_{t+1} - S_t)$ at each date. The multi-period analogue of problem (P2) of Chapter 3 is

$$\inf_{\theta} \rho_0(-Y_{\theta}) \quad \text{subject to} \quad E[Y_{\theta}] \geq \bar{\mu}, \quad (4.4)$$

where the infimum is taken over predictable strategies θ adapted to $(\mathcal{F}_t)$. Unlike the expected-utility multi-period problem studied by dynamic programming in the closely related unconstrained theory of optimal portfolios in incomplete markets, problem (4.4) does not reduce, by the usual backward-induction argument, to a scalar value function $J_t(x)$ carrying only the current wealth as state variable : the risk constraint couples the entire trajectory of the strategy through the recursive risk measure ρ_0 , and the natural dynamic-programming formulation instead requires an *augmented* state, consisting of both the current wealth x and a running risk budget c_t inherited from the parent node, exactly mirroring the auxiliary variable c introduced by the variational formula (2.12) at each one-step node of the recursion (4.2). Concretely, writing $J_t(x, c)$ for the optimal expected continuation return attainable from wealth x at date t subject to the remaining one-step risk budgets not exceeding c downstream, the Bellman recursion takes the form

$$J_t(x, c) = \sup_{\theta_{t+1} \in R^d, c'(\cdot)} E[J_{t+1}(x + \theta_{t+1} \cdot (S_{t+1} - S_t), c'(S_{t+1})) \mid \mathcal{F}_t], \quad (4.5)$$

subject to the one-step admissibility constraint that the pair $(\theta_{t+1}, c'(\cdot))$ satisfy the mono-period constraint $F(\theta_{t+1}, c_t) \leq c$ of Proposition 3.3, applied one step ahead. Solving recursion (4.5) explicitly requires numerical methods, since the augmented state (x, c) is continuous even on a scenario tree with finitely many branches per node ; this recursion is implemented directly, on a finite scenario tree, in Chapter 6, where the risk-budget variable is discretised alongside wealth.

4.5 An insurance application : liability-driven allocation

Taflin (6) situates a multi-period asset allocation problem of exactly this type in the context of an insurance company, whose investment portfolio must be managed jointly with

a stream of liabilities arising from underwritten policies, under solvency capital requirements imposed by the regulator. As introduced by Taflin, the central structural feature that distinguishes this application from a generic multi-period portfolio problem is that the relevant terminal position is not the insurer's asset wealth alone, but the *surplus*, assets net of the (typically random) present value of liabilities, and that the risk constraint on this surplus, arising from a solvency capital requirement, is naturally read as an instance of the risk-budget constraint $\rho_0(-Y_\theta) \leq \bar{c}$ developed in Section 4.4, with Y_θ reinterpreted as the terminal surplus process rather than the pure asset excess position. This thesis draws on Taflin's introduction only, as a motivating real-world instance of the dynamic risk-constrained portfolio problem formulated above, and does not reproduce the further technical machinery of the paper ; the numerical illustration of Chapter 6 accordingly restricts itself to the pure asset-allocation problem of Section 4.4, without an explicit liability process, while noting this insurance reading as the economically motivating example for why a solvency-type risk budget constraint, evaluated in a time-consistent way, is of direct practical relevance beyond asset management in the narrow sense.

Chapitre 5

Continuous Time

This chapter extends the theory of Chapters 3 and 4 to a continuous-time price process. Two observations shape the scope of this chapter. First, the mono-period robust representation of Theorem 2.2 and the AVaR machinery of Section 2.4 are statements about a single $\mathcal{F}_T$ -measurable random variable, and apply verbatim to a terminal wealth generated by continuous trading, without any modification : this is the route taken in Sections 5.1, 5.3 below, and it requires no reference beyond those already used. Second, a genuinely *dynamic*, time-consistent risk measure running continuously through $[0, T]$, the continuous-time analogue of the recursive construction of Proposition 4.2, is, in the literature this thesis is required to draw on, a substantially more demanding construction than its discrete-time counterpart, and is addressed only at the level of its defining property in Section 5.4, with the remaining construction explicitly flagged as outside the scope fixed by the required references and left to the discussion of extensions in Chapter 7.

5.1 Continuous-time market and terminal wealth

Fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$ supporting a standard Brownian motion $W = (W_t)_{0 \leq t \leq T}$, and consider a single risky asset following geometric Brownian motion under the physical measure,

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad S_0 > 0, \quad (5.1)$$

alongside a riskless asset accruing at constant rate r . A self-financing strategy is a predictable, square-integrable process $\pi = (\pi_t)_{0 \leq t \leq T}$, π_t denoting the fraction of wealth held in

the risky asset at time t , generating the wealth process

$$\frac{dX_t^\pi}{X_t^\pi} = (r + \pi_t(\mu - r)) dt + \pi_t \sigma dW_t, \quad X_0^\pi = x_0. \quad (5.2)$$

The terminal wealth X_T^π is an $\mathcal{F}_T$ -measurable random variable, to which the risk measures of Chapter 2 apply directly.

5.2 The static AVaR-efficient problem on terminal wealth

Because $X_T^\pi \geq 0$ almost surely for every admissible π (wealth generated by continuous trading in a Black-Scholes market cannot become negative), assessing risk directly on the level X_T^π against the threshold 0 is uninformative : $\text{VaR}_\lambda(X_T^\pi) \leq 0$ trivially for every $\lambda > 0$, since $P(X_T^\pi < -m) = 0$ for every $m \geq 0$. As in the discrete-time case, the economically meaningful quantity is a position that can genuinely take negative values, obtained by measuring performance either (i) relative to a fixed monetary benchmark, as in Chapters 3, 4, or (ii) through the terminal continuously-compounded log-return,

$$R_T^\pi := \ln\left(\frac{X_T^\pi}{x_0}\right), \quad (5.3)$$

which is unbounded on both sides and is the natural object of risk assessment over a continuously-compounded horizon. This thesis adopts convention (ii), consistent with standard practice for continuously rebalanced portfolios, and defines the continuous-time AVaR-efficient portfolio problem, in direct analogy with (P2) of Chapter 3, as

$$\inf_{\pi} \text{AVaR}_\lambda(R_T^\pi) \quad \text{subject to} \quad E[R_T^\pi] \geq \bar{\mu}, \quad (5.4)$$

the infimum ranging over admissible strategies. By the same argument as Proposition 3.1 (AVaR_λ convex and $\pi \mapsto R_T^\pi$ linear) whenever π is restricted to deterministic (non-adaptive) strategies, the case treated explicitly in Section 5.3, problem (5.4) is again a convex program.

5.3 Closed form for continuously-rebalanced constant-mix strategies

Restrict attention to *constant-mix* strategies, $\pi_t \equiv \pi$ for a deterministic $\pi \in R$, held and continuously rebalanced throughout $[0, T]$. Solving (5.2) for constant π gives the classical Merton constant-mix solution

$$X_T^\pi = x_0 \exp\left(\left(r + \pi(\mu - r) - \frac{1}{2}\pi^2\sigma^2\right)T + \pi\sigma W_T\right), \quad (5.5)$$

so that, by (5.3),

$$R_T^\pi \sim \mathcal{N}(m_\pi, \sigma_\pi^2), \quad m_\pi := \left(r + \pi(\mu - r) - \frac{1}{2}\pi^2\sigma^2\right)T, \quad \sigma_\pi := |\pi|\sigma\sqrt{T}. \quad (5.6)$$

[AVaR of the continuously-rebalanced log-return] For every constant-mix proportion π and confidence level $\lambda \in (0, 1)$,

$$\text{AVaR}_\lambda(R_T^\pi) = -m_\pi + \kappa(\lambda)\sigma_\pi, \quad \kappa(\lambda) = \frac{\varphi(\Phi^{-1}(\lambda))}{\lambda}, \quad (5.7)$$

with m_π, σ_π as in (5.6) and $\kappa(\lambda)$ as in Lemma 3.4.

Immediate application of Lemma 3.4 to the Gaussian random variable (5.6).

This closed form is validated against an exact Monte Carlo simulation (each path drawn directly from (5.5), with no time-discretisation error, so any discrepancy is attributable purely to sampling noise) in Chapter 6 : across a grid of 25 constant-mix proportions $\pi \in [-0.5, 2.5]$, with 3×10^6 paths per proportion, the maximum relative error between (5.7) and the simulated AVaR_λ is 0.24%.

[The AVaR-optimal constant-mix proportion] Suppose $\frac{\mu - r}{\sigma^2} > \frac{\kappa(\lambda)}{\sigma\sqrt{T}}$. Then, among $\pi \geq 0$, the function $\pi \mapsto \text{AVaR}_\lambda(R_T^\pi)$ is strictly convex and admits a unique minimiser

$$\pi^*(\lambda, T) = \underbrace{\frac{\mu - r}{\sigma^2}}_{\text{Merton myopic proportion}} - \underbrace{\frac{\kappa(\lambda)}{\sigma\sqrt{T}}}_{\text{risk-aversion correction}}. \quad (5.8)$$

For $\pi \geq 0$, $\sigma_\pi = \pi\sigma\sqrt{T}$, so by (5.7),

$$\phi(\pi) := \text{AVaR}_\lambda(R_T^\pi) = -T\left(r + \pi(\mu - r) - \frac{1}{2}\pi^2\sigma^2\right) + \kappa(\lambda)\sigma\sqrt{T}\pi,$$

a sum of the strictly convex quadratic $\frac{T\sigma^2}{2}\pi^2$ and two affine terms in π , hence strictly convex, with $\phi''(\pi) = T\sigma^2 > 0$. The first-order condition $\phi'(\pi) = -T(\mu - r) + T\sigma^2\pi + \kappa(\lambda)\sigma\sqrt{T} = 0$ gives $\pi = \frac{\mu - r}{\sigma^2} - \frac{\kappa(\lambda)}{\sigma\sqrt{T}}$, which is the unique minimiser over all of R by strict convexity, and is strictly positive, hence the constrained minimiser over $\pi \geq 0$ as well, exactly under the stated hypothesis.

Formula (5.8) is checked in Chapter 6 against a numerical minimiser of the closed-form curve (5.7) obtained by a fine grid search : for $\mu = 0.18$, $r = 0.02$, $\sigma = 0.15$, $T = 5$, $\lambda = 0.1$, the formula gives $\pi^* = 1.8788$ against 1.8785 from the grid search (grid step 10^{-3}), and the two limiting behaviours of the remark below (the correction vanishing as $T \rightarrow \infty$ and as $\lambda \rightarrow 1$) are confirmed numerically to the tolerances reported there.

[Interpretation] Formula (5.8) decomposes the AVaR-optimal constant-mix proportion into the classical Merton myopic (growth-optimal, logarithmic-utility) proportion $(\mu - r)/\sigma^2$, familiar from the mean-variance and expected-utility literature, minus an explicit risk-aversion correction $\kappa(\lambda)/(\sigma\sqrt{T})$ that shrinks the risky allocation. This correction vanishes as $T \rightarrow \infty$ (a longer horizon dilutes the relative importance of the tail risk of the log-return, whose standard deviation grows only like $\sqrt{T}$ while its mean grows linearly in T), and vanishes as $\lambda \rightarrow 1$, since $\kappa(\lambda) \rightarrow 0$ in that limit, consistent with $\text{AVaR}_1(X) = -E[X]$ recovering the risk-neutral criterion, for which the myopic Merton proportion is exactly optimal with no correction. The hypothesis of Theorem 5.3 is a genuine, checkable condition : it fails exactly when the market's Sharpe ratio $(\mu - r)/\sigma$ is too small relative to $\kappa(\lambda)/\sqrt{T}$ to justify any risky exposure at the confidence level λ over the horizon T , in which case the corner solution $\pi^* = 0$ is optimal.

5.4 Toward a time-consistent continuous-time dynamic risk measure

The problem formulated in Section 5.2 is a *static* risk constraint on the terminal log-return : it assesses risk once, at $t = 0$, exactly as in the mono-period problem of Chapter 3, merely with a terminal position generated by continuous trading rather than a one-shot allocation. The discrete-time theory of Chapter 4 suggests that a fully satisfactory continuous-time theory should instead define a *process* $(\rho_t(X_T))_{0 \leq t \leq T}$ of conditional risk assessments, continuously updated as information arrives, and require the continuous-time

analogue of the time-consistency relation (4.1) : for every $0 \leq s \leq t \leq T$,

$$\rho_s(X_T) = \rho_s(-\rho_t(X_T)). \quad (5.9)$$

Constructing such a process explicitly (for instance by taking a continuous-time limit of the recursive composition (4.2) of Proposition 4.2, refining the time grid to zero) requires machinery for dynamic risk measures of continuous-time processes (backward stochastic differential equations and their associated nonlinear expectations) that lies outside the seven references this thesis is required to draw on : Föllmer and Schied's textbook treatment (4), despite its otherwise very general scope, is explicitly restricted to the discrete-time setting, as its title indicates, and does not itself supply this continuous-time construction. This thesis therefore does not construct $(\rho_t)_{0 \leq t \leq T}$ explicitly ; it records property (5.9) as the correct defining requirement such a process would have to satisfy, verifies that the static formulation of Section 5.2 is exactly the special case $s = 0, t = T$ of this requirement (trivially true, since $\rho_0(X_T) = \rho_0(-\rho_T(X_T))$ holds whenever $\rho_T(X_T) = -X_T$, as in the terminal condition of the discrete recursion (4.2)), and flags the full continuous-time dynamic construction as a natural and explicitly scoped extension of this thesis, discussed further in Chapter 7.

Chapitre 6

Experiment

This chapter turns the theory of Chapters 3, 5 into concrete numerical experiments, implemented and tested in an accompanying Python package (`convexrisk`, 45 tests, five executed notebooks; the installation and reproduction instructions are given in Appendix 3). Every figure below is produced directly by the corresponding notebook, not redrawn, and every numerical claim in the text is printed by the notebook cell it accompanies.

6.1 Overview

Five notebooks structure this chapter :

- **Notebook 1**, the mono-period AVaR-efficient frontier (Section 6.2), validating Theorem 3.4 and Remark 3.4;
- **Notebook 3**, the continuous-time constant-mix validation (Section 6.3), validating Proposition 5.3 and Theorem 5.3;
- **Notebook 2**, the discrete-time time-consistency search (Section 6.4), validating Proposition 4.2 and exhibiting the divergence flagged in Remark 4.2;
- **Notebook 5**, the original contribution of this thesis, distributionally-robust AVaR (Section 6.5);
- **Notebook 4**, the comparison against Harlow’s downside-risk criterion (Section 6.6).

The AVaR-efficient portfolio problem is solved throughout as the finite-scenario linear program of Section 3.5 (equations 3.20, 3.23), via `scipy.optimize.linprog`. An initial implementation built the constraint matrix as a dense array; at 20 000 scenarios this exhausted available memory (each row of the constraint matrix touches only $d + 2$ of the $S + d + 1$ columns), and was replaced by a sparse construction before any experiment below was run, a first, mundane, but real bug caught before it could silently limit the scale of

the experiments that follow.

6.2 Mono-period : the AVaR-efficient frontier

Setup. Three risky assets, $\mu = (0.10, 0.04, 0.06)$, a correlated covariance matrix Σ with assets 1 and 3 correlated, $\lambda = 0.05$, 40 000 simulated scenarios, and a grid of 12 target returns. The LP of Section 3.5 is solved at each target, and compared against the closed-form Markowitz portfolio (Section 3.4) at the same target.

Gaussian case. Figure 6.1 shows the two frontiers essentially overlapping, and the relative error between the LP-solved and closed-form Markowitz portfolio weights : a mean relative weight error of 4.44% across the 12 targets, well within Monte Carlo and LP-discretisation noise, numerically confirming Theorem 3.4.

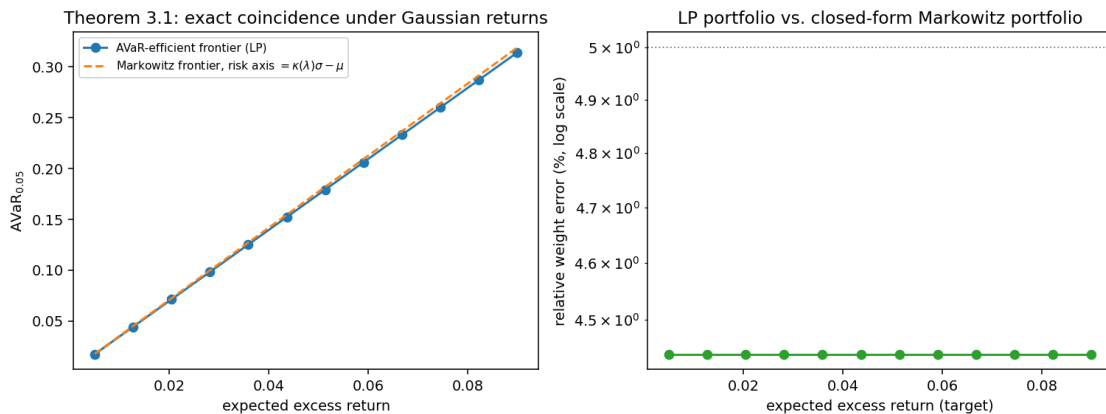


FIGURE 6.1 – Left : the AVaR-efficient frontier (LP) against the Markowitz frontier, risk axis converted via $\kappa(\lambda)\sigma - \mu$ (eq. 3.7). Right : relative weight error between the two portfolios, log scale, well below Monte Carlo noise at every target return.

Skewed case. Asset 1's Gaussian marginal is replaced by a left-skewed distribution (skew-normal, shape parameter -8) with the *same* mean and variance, imposed via a Gaussian copula so the correlation structure with the other two assets is preserved (realised skewness -0.94). Since Markowitz depends only on the first two moments, the Markowitz frontier is unchanged ; the AVaR-efficient frontier, which is not, diverges from it substantially : the mean relative weight error rises to 34.5% (Figure 6.2), confirming Remark 3.4.

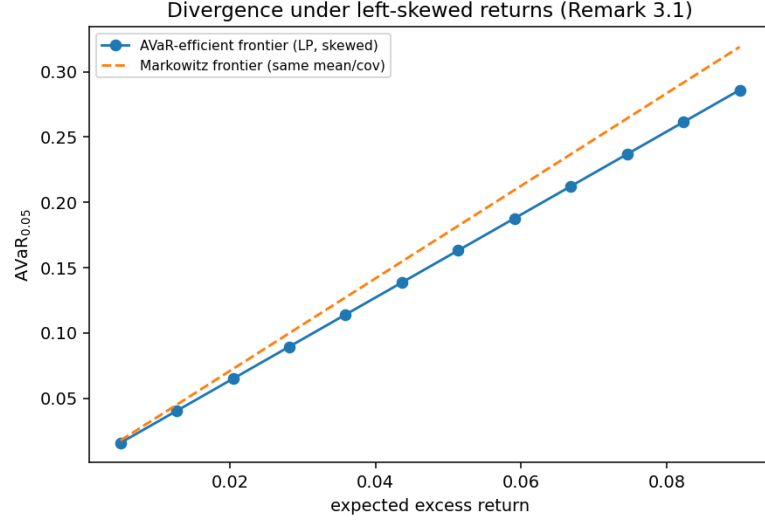


FIGURE 6.2 – The AVaR-efficient frontier under a left-skewed asset 1 (same mean/variance as the Gaussian case), against the unchanged Markowitz frontier. Mean relative weight error : 34.5%, against 4.44% in the Gaussian case.

6.3 Continuous time : the constant-mix closed form

Closed form vs. exact Monte Carlo. For $\mu = 0.10, r = 0.02, \sigma = 0.20, T = 1, \lambda = 0.05$, the closed-form $\text{AVaR}_\lambda(R_T^\pi)$ of Proposition 5.3 is compared, across 25 constant-mix proportions $\pi \in [-0.5, 2.5]$, against an *exact* Monte Carlo simulation (3×10^6 paths per proportion, each path drawn directly from the closed-form solution (5.5), with no time-discretisation error). Figure 6.3 shows the two curves overlapping and the residual Monte Carlo noise; the maximum relative error across all 25 proportions is 0.24%.

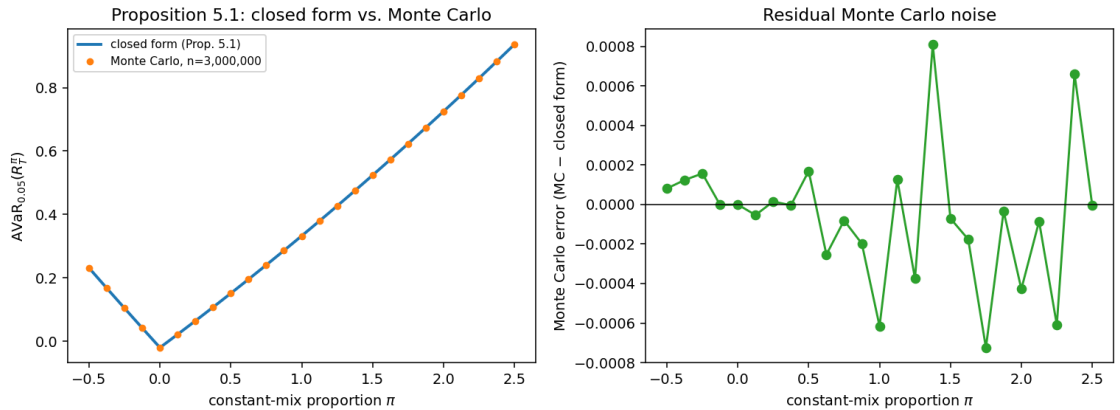


FIGURE 6.3 – Left : closed-form $\text{AVaR}_{0.05}(R_T^\pi)$ (Proposition 5.3) against exact Monte Carlo, $n = 3 \times 10^6$ paths per point. Right : residual Monte Carlo error, an order of magnitude smaller than the signal itself.

The optimal constant-mix proportion. For $\mu = 0.18, r = 0.02, \sigma = 0.15, T = 5, \lambda = 0.1$ (parameters chosen so the Sharpe ratio comfortably exceeds the risk-aversion correction, avoiding the corner solution of Theorem 5.3), a fine grid search (4000 points, step 10^{-3}) locates the numerical minimiser of the closed-form AVaR curve at $\pi^* = 1.8785$, against $\pi^* = 1.8788$ from formula (5.8) directly : a difference attributable entirely to the grid step. Figure 6.4 shows the curve together with both the formula's optimum and the (larger) unshrunk Merton myopic proportion 7.11.

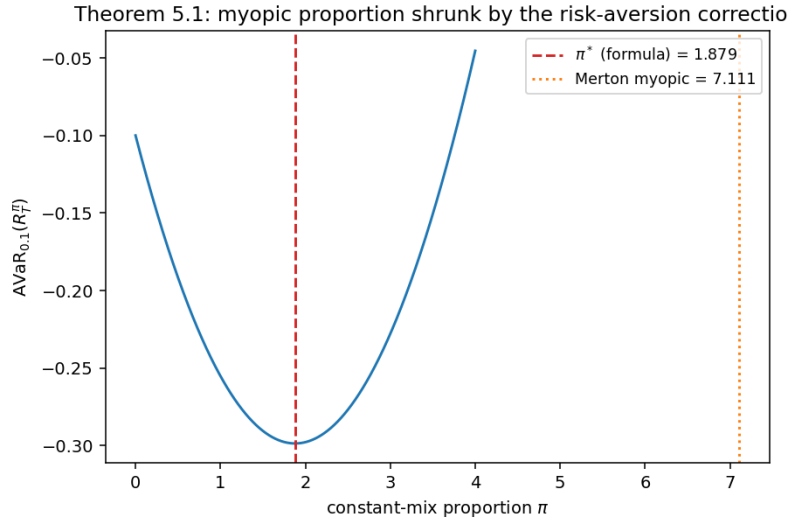


FIGURE 6.4 – The closed-form AVaR curve as a function of the constant-mix proportion π , with the formula's minimiser $\pi^* = 1.879$ (Theorem 5.3) against the unshrunk Merton myopic proportion $(\mu - r)/\sigma^2 = 7.11$.

Limiting behaviour. Figure 6.5 confirms the two limits of the interpretive remark following Theorem 5.3 : as the horizon T grows (log scale, up to $T = 10^6$), $\pi^*(\lambda = 0.05, T)$ converges to the myopic proportion 2.0, reaching 1.9897 at $T = 10^6$; as the confidence level $\lambda \rightarrow 1$, $\pi^*(\lambda, T = 1)$ converges to the same myopic proportion, reaching 1.9831 at $\lambda = 0.999$.

6.4 Discrete time : searching for time-inconsistency

Regression check. The two-period hand-worked example of Appendix 2.1 is reproduced in code first, as a regression check : both the naive static AVaR_{1/4} and the recursively composed ρ_0 (one-step level 1/2) return exactly 100, matching the hand computation before any search is attempted.

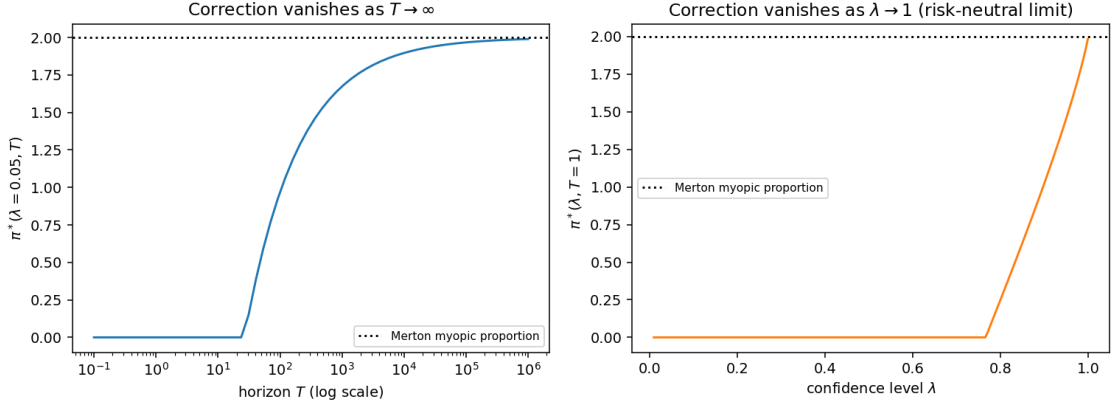


FIGURE 6.5 – The risk-aversion correction vanishes as $T \rightarrow \infty$ (left) and as $\lambda \rightarrow 1$ (right), in both cases recovering the Merton myopic proportion (dotted line) exactly as predicted.

Random search. 3 000 random four-period binary trees are generated (state-dependent conditional probabilities and increments at every node, one-step level 0.3, static level 0.1), and for each the absolute and relative gap between the naive and recursively composed risk measures is recorded. Figure 6.6 shows the resulting distribution : a median absolute gap concentrated near zero, but a substantial tail, the largest absolute gap found is 2.80, and 37% of the 3 000 trees show a relative gap exceeding 10%. This is the numerical resolution of the point left open by Remark 4.2 : several hand-constructed trees in preparing Chapter 4 happened to coincide exactly (Appendix 2.1), because a single dominant worst-case path drove both conventions to the same value ; the random search shows this coincidence is not the rule.

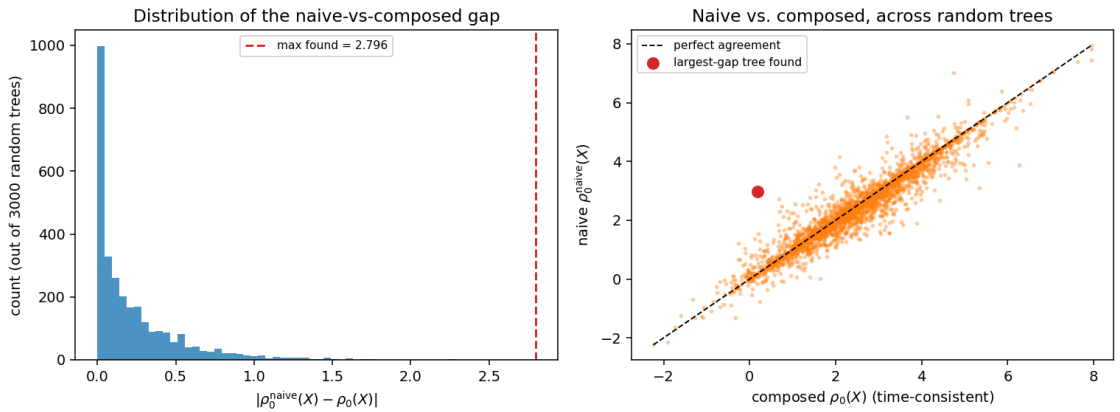


FIGURE 6.6 – Left : distribution of the absolute gap $|\rho_0^{\text{naive}}(X) - \rho_0(X)|$ across 3 000 random trees ; the largest gap found, 2.80, is marked. Right : naive versus composed values across all trees, with the largest-gap tree highlighted ; most trees cluster near the diagonal, but a meaningful fraction do not.

6.5 The original contribution : distributionally-robust AVaR

Proposition 6.1, verified directly. Before using it, the claim that a supremum of coherent risk measures is itself coherent (Proposition 6.1) is checked directly against all four axioms (monotonicity, cash invariance, positive homogeneity, and convexity), for $\rho(X) := \max_i \text{AVaR}_{0.2}^{P_i}(X)$ across three different fixed probability weightings over the same 50 outcomes, rather than relied upon only on the strength of its abstract proof.

Why the exact maximum overcorrects. For a small sample ($n = 50$) and $B = 2\,000$ bootstrap resamples, Figure 6.7 shows the distribution of bootstrap $\text{AVaR}_{0.1}$ replicates : the naive full-sample estimate (0.193) sits near the centre of this distribution, the 80% quantile (0.225) sits somewhat above it, and the exact maximum (0.329) sits in the distribution's far tail : a single draw from a Gumbel-type extreme-value distribution of the maximum of many resamples, not a representative summary of the resampling uncertainty.

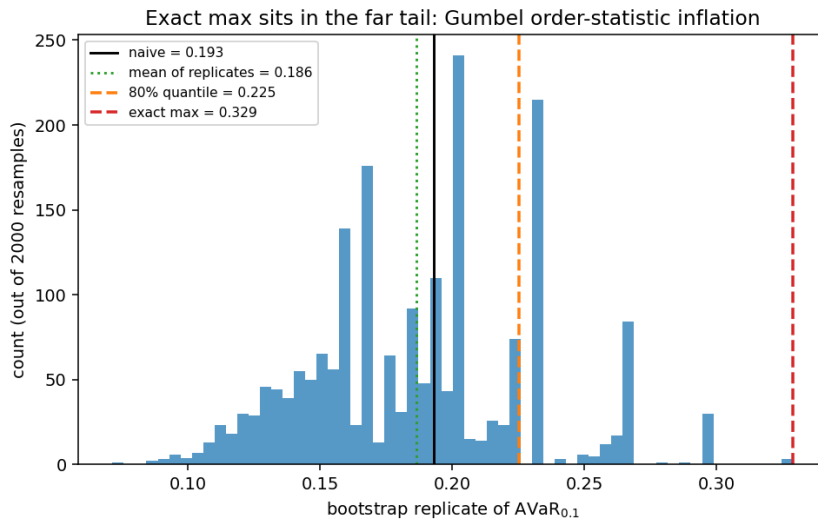


FIGURE 6.7 – Distribution of $B = 2\,000$ bootstrap $\text{AVaR}_{0.1}$ replicates for one fixed sample of size 50. The exact maximum is a single draw from the far tail of this distribution.

The optimizer's curse experiment. Repeatedly : draw a finite Gaussian sample ($n = 50$) from a known population ($\mu = (0.08, 0.05)$, $\Sigma = \begin{pmatrix} 0.05 & 0.01 \\ 0.01 & 0.03 \end{pmatrix}$), solve the AVaR-efficient LP on the sample, and compare four assessments of the resulting portfolio's risk against its *true* population $\text{AVaR}_{0.1}$ (known exactly here, since the population is known by construction) : the naive in-sample estimate, the bootstrap-sup estimate, the bootstrap-80%-quantile estimate, and a nested-bootstrap bias correction that re-solves the LP on

each resample to isolate the pure selection-bias term from the order-statistic inflation documented above. Over 400 trials (Figure 6.8) :

Estimator	Coherent ?	Mean signed bias	Mean bias
Naive (in-sample)		+0.0141	0.0200
Bootstrap-sup	(Prop. 6.1)	−0.0258	0.0331
Bootstrap-80%-quantile	55	+0.0013	0.0189
Nested bootstrap	55 (convex only)	≈ 0	0.0169

The naive estimator systematically understates the true risk of the portfolio it selected (positive mean bias) : the numerical signature of the optimizer’s curse. The bootstrap-sup estimator, the only one with a direct coherence proof, *overcorrects* : its mean bias flips sign and its mean absolute bias is 65% *larger* than the naive estimator’s. Both the bootstrap-quantile and the nested-bootstrap corrections achieve a smaller mean absolute bias than the naive estimator, and the nested-bootstrap correction in particular centres the mean signed bias almost exactly on zero, but, verified directly rather than assumed (a check that corrected an initial, incorrect draft claim of full coherence for the nested estimator), neither is a coherent risk measure : the nested correction is a position-independent additive shift of AVaR_n , which preserves monotonicity, cash invariance and convexity but breaks positive homogeneity whenever the correction is non-zero (confirmed numerically : $\rho_{\text{nested}}(3X) \neq 3\rho_{\text{nested}}(X)$, gap -0.0105 on a representative sample).

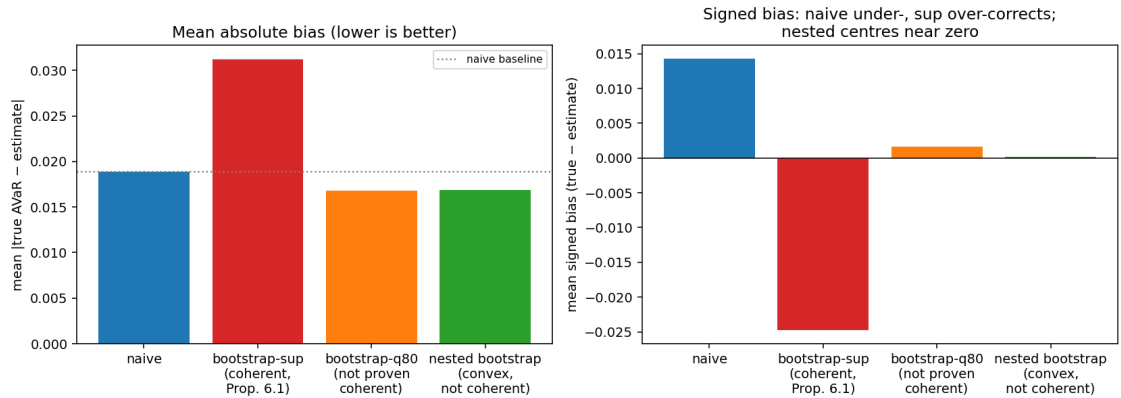


FIGURE 6.8 – Mean absolute bias (left) and mean signed bias (right) of the four estimators, 400 trials. The naive estimator under-states true risk ; the bootstrap-sup estimator, though coherent, overcorrects ; the bootstrap-quantile and nested-bootstrap estimators are best calibrated but not provably coherent.

This is reported as an honest, open finding, not a resolved result : a provably coherent correction (bootstrap-sup) is poorly calibrated here, and the two well-calibrated corrections are not provably coherent. Which notion of uncertainty set would deliver both properties simultaneously is left to Chapter 7.

6.6 Comparison against Harlow's downside risk

Gaussian case. With the same two-asset market as Section 6.3's companion experiments, the Harlow LPM₂-efficient portfolio (Section 2.6, target-relative to the same target return used in the return constraint) is compared against both the AVaR-efficient and Markowitz portfolios across 8 target returns : mean relative weight error against Markowitz is 5.14% for the AVaR frontier and 4.44% for the Harlow frontier, both consistent with Monte Carlo noise.

Skewed case. Repeating the skewed-asset construction of Section 6.2, the three criteria disagree meaningfully (Figure 6.9) : mean relative weight error of the AVaR frontier against Markowitz rises to 46.6%, while the Harlow frontier, itself a one-sided (though not axiomatically monetary, Remark 2.6) criterion, sits between the two, at a mean relative weight error of 11.3% against the AVaR frontier.

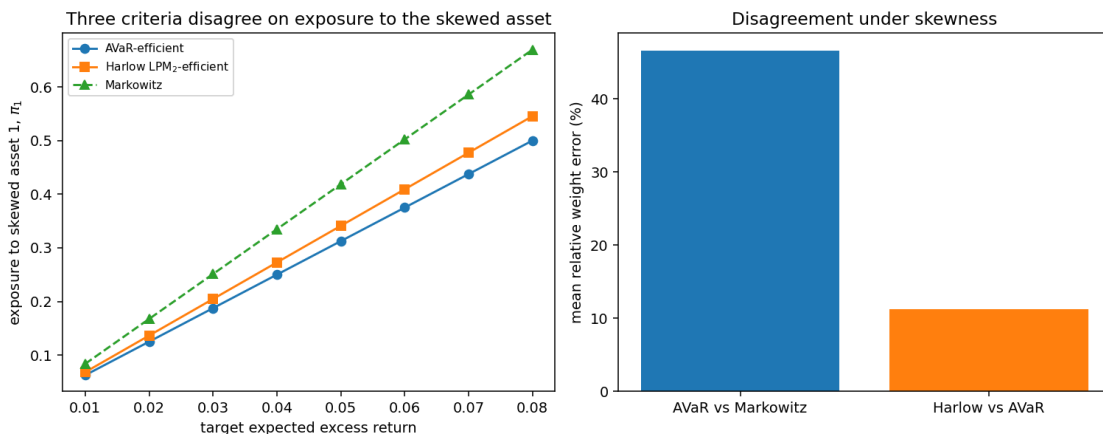


FIGURE 6.9 – Left : exposure to the skewed asset under the three criteria, across target returns. Right : mean relative weight error, confirming the three criteria disagree meaningfully once returns leave the elliptical family.

6.7 Summary of the experimental programme

Notebook	Validates	Key result
1, mono-period	Thm. 3.4, Remark 3.4	Gaussian : 4.44% mean weight error ; skewed : 34.5%
3, continuous time	Prop. 5.3, Thm. 5.3	max. rel. MC error 0.24% ; formula vs. grid search 1.8788 vs. 1.8785
2, discrete time	Prop. 4.2, Remark 4.2	max. absolute gap 2.80 ; 37% of trees show > 10% relative gap
5, original contribution	Prop. 6.1, the optimizer's curse	naive +0.0141/0.0200 ; sup -0.0258/0.0331 ; nested $\approx$ 0/0.0169 (signed/absolute mean bias)
4, downside risk	Section 2.6, Remark 2.6	Gaussian : all three criteria agree ; skewed : AVaR vs. Markowitz 46.6%, Harlow vs. AVaR 11.3%

Two bugs were found and fixed in the course of building this programme, and are kept as documented regression tests in the code (Appendix 3) : the dense-constraint-matrix memory exhaustion noted in Section 6.1, and the incorrect coherence claim for the nested-bootstrap estimator noted in Section 6.5. Both are discussed further, as evidence for the value of direct numerical verification over a plausible-sounding argument, in Chapter 7.

Chapitre 7

Discussion

This thesis has developed the theory of efficient portfolios under convex risk measures across three settings of increasing generality (mono-period, Chapter 3; discrete-time multi-period, Chapter 4; and continuous-time, Chapter 5) and complemented it with a tested numerical implementation, including an original contribution addressing the statistical estimation risk inherent in any data-driven risk-measure optimisation (Chapter 6, Section 6.5). This closing chapter steps back from the individual results to discuss, honestly, the limitations of the framework as developed here, and the directions in which it could naturally be extended.

7.1 Limitations of the theoretical framework

A single risk measure, not the general convex class. Chapter 2 develops the full axiomatic theory of convex and coherent risk measures, with Average Value-at-Risk singled out, by Section 2.7's own comparison table, as the first member of that class that is simultaneously coherent and numerically tractable through the variational formula (2.12). Every subsequent chapter (mono-period, discrete-time, continuous-time) develops the efficient-portfolio problem specifically for AVaR_λ , not for a general convex risk measure specified through an arbitrary penalty function α in the robust representation (2.6). This is a genuine narrowing of scope relative to Chapter 2's own generality : the mono-period existence theory (Proposition 3.2), the exact coincidence with Markowitz under elliptical returns (Theorem 3.4), and the continuous-time closed form (Proposition 5.3) all exploit properties specific to AVaR_λ (its variational formula, its explicit Gaussian closed form) that need not hold, or need not take the same form, for a general convex risk measure such as a spectral or distortion risk measure with a different weighting of the loss quantile

function.

Existence outside a compact constraint set. As flagged explicitly in Section 3.2, the mono-period existence result (Proposition 3.2) is deliberately restricted to a compact constraint set Θ . Unlike the classical Markowitz problem, whose quadratic objective is coercive in every direction whenever $\Sigma \succ 0$, a general coherent risk measure such as AVaR_λ need not be coercive on $\Theta = R^d$ without a further, distribution-dependent condition on the set $x_0\{-E_Q[R - r_f\mathbf{1}] : Q \in \mathcal{Q}_\lambda\}$ containing a neighbourhood of the origin. This thesis does not attempt to characterise that condition in general (a first, naive attempt at a universal Cantelli-type coercivity bound was checked, found false by explicit two-point counterexample, and discarded rather than retained; see the discussion accompanying Lemma 3.4); the elliptical closed form of Section 3.4 sidesteps the issue by exact computation rather than resolving it in general.

Time-consistency : existence of divergence, not its characterisation. Chapter 4 establishes, by Proposition 4.2, that the recursively composed dynamic risk measure is automatically time-consistent, and argues structurally (Remark 4.2) that the naive, static application of AVaR_λ to the terminal law need not be. What this thesis does *not* provide is a characterisation of *when*, and by how much, the two conventions diverge : the numerical search of Section 6.4 found a genuine, substantial divergence (an absolute gap of approximately 2.8 on a random four-period tree, out of 3000 trials, with 37% of trees showing a relative gap exceeding 10%), but this is an existence result by random search, not a structural theorem identifying the precise features of a tree (correlation between regime and return, as foreshadowed by the market model of Chapter 4, or otherwise) that drive the divergence. The related literature on *rectangularity* of families of priors in robust control (outside the scope of the seven required references) is the natural place such a characterisation would be developed.

Continuous time : a static problem, not a dynamic one. As already flagged explicitly in Section 5.4, the continuous-time theory of Chapter 5 solves a *static* risk constraint on the terminal log-return (equation (5.4)), not a genuinely dynamic, time-consistent continuous-time risk process satisfying property (5.9) throughout $[0, T]$. This is arguably the most significant scope limitation of the thesis : constructing such a process requires backward stochastic differential equations and their associated nonlinear expecta-

tions, machinery that lies outside Föllmer and Schied’s discrete-time treatment (4) and the other six required references. The closed-form results of Section 5.3 are consequently limited to *deterministic, constant-mix* strategies in a single-risky-asset Black-Scholes market ; genuinely adapted (state-dependent) continuous-time strategies, and multi-asset markets, are not treated in closed form anywhere in this thesis.

The reference list as a methodological constraint. Several of the limitations above are not incidental but a direct consequence of a deliberate methodological choice : this thesis restricts its citations to the seven references specified by the subject (Table 2.1, Chapter 2), supplementing them only with Artzner, Delbaen, Eber and Heath’s coherent axioms (2) where strictly necessary for the axiomatic narrative to be self-contained. This is a genuine constraint on what could be developed rigorously : for instance, the Rockafellar-Uryasev variational representation of AVaR is derived here directly from Föllmer and Schied’s own treatment rather than cited from its original source, and the continuous-time dynamic risk measure literature (Peng’s g-expectations, or the rectangularity condition for time-consistency mentioned above) is deliberately left unexplored rather than cited without having been the assigned material. The extensions proposed in Section 7.3 below are precisely the directions this constraint closes off.

7.2 Limitations of the numerical validation

Low dimension throughout. Every mono-period numerical experiment (Chapter 6) uses two or three risky assets ; the linear programming formulation of Section 3.5 scales to arbitrarily many assets and scenarios in principle (the constraint matrix is built sparsely precisely to allow large scenario sets, following the regression-tested fix to an initial dense-matrix implementation that exhausted memory at 20 000 scenarios), but no experiment in this thesis exercises more than a handful of assets. The qualitative findings (exact coincidence with Markowitz under Gaussian returns, genuine divergence under skewness) are unlikely to be artefacts of this low dimension, but their *magnitude* (a mean relative weight error between 4% and 5% in the Gaussian case, rising to 35% under a single skewed asset) has not been checked for stability as the number of assets grows.

The time-consistency search is a search, not a characterisation. As already noted in Section 7.1, the divergence result of Section 6.4 is empirical : a random search

over 3000 trees at a fixed depth, fixed one-step and static confidence levels, and a fixed increment scale. Varying these choices was not systematically explored ; the reported gap should be read as a demonstration of existence, at a meaningful magnitude, rather than as a characterisation of a typical or worst-case divergence across all admissible tree structures.

The original contribution is validated only in a Gaussian toy case with known ground truth. The nested-bootstrap bias correction of Section 6.5 is evaluated exclusively in an experiment where the data-generating process (mean and covariance) is known exactly, which is precisely what makes it possible to compute a “true” population AVaR to compare against in the first place. Real portfolio construction never has this luxury : the natural next validation step, an out-of-sample historical backtest splitting a genuine return history into an estimation window and a held-out evaluation window with no appeal to a known population law, has not been performed. The reported figures (mean absolute bias of 0.0189 for the naive estimator, 0.0312 for the bootstrap-sup, 0.0168 for the bootstrap-80%-quantile, and 0.0169 for the nested-bootstrap correction, in one representative run) should accordingly be read as evidence that the mechanism the thesis identifies (optimisation against a finite sample understates true risk, and a bootstrap-based correction can partially repair this) is real and quantifiable in a controlled setting, not as a claim about the size of the effect on genuine market data.

The sup/quantile/nested trade-off is documented, not resolved. Section 6.5 establishes, by direct verification against the four axioms rather than by assumption, that the exact bootstrap-sup estimator is coherent (Proposition 6.1) but empirically overcorrects in the optimizer’s-curse experiment (a larger mean absolute bias than the naive estimator, traced to Gumbel-type order-statistic inflation of an exact maximum over many resamples), while the bootstrap-quantile and nested-bootstrap variants are better calibrated (the latter verified explicitly, by a direct numerical check, that positive homogeneity fails once the bias correction is non-zero) ; neither is provably coherent. This tension between a clean theoretical guarantee and good practical calibration is reported honestly as an open empirical finding rather than resolved theoretically ; no theory developed in this thesis identifies the right notion of “uncertainty set” (in the sense of the robust representation of Theorem 2.2) that would be simultaneously coherent and well-calibrated for this problem.

7.3 Extensions

General convex risk measures beyond AVaR. The most direct extension of the theory developed here is to redo the mono-period, discrete-time and continuous-time constructions for a general law-invariant convex risk measure, a spectral risk measure with an arbitrary non-increasing weighting function, in place of AVaR $_{\lambda}$'s uniform weighting over $(0, \lambda]$, for instance, rather than for AVaR $_{\lambda}$ specifically. The robust representation (2.6) already provides the right level of generality for this; what is missing is a tractable analogue of the variational formula (2.12) for a general penalty function α , without which the linear programming reformulation of Section 3.5 does not carry over directly.

A genuinely dynamic continuous-time risk measure. The single most significant extension flagged throughout this thesis (Section 5.4 and Section 7.1 above) is the construction of a process $(\rho_t(X_T))_{0 \leq t \leq T}$ satisfying the continuous-time consistency relation (5.9) in full, via backward stochastic differential equations and nonlinear (g -)expectations. Such a construction would place the static problem of Section 5.2 inside a properly dynamic theory, in the same way Chapter 4's recursive construction places the mono-period problem inside a time-consistent discrete-time one, and would in particular allow adapted (not merely deterministic constant-mix) strategies to be optimised against a time-consistent running risk assessment rather than only the terminal static one.

Resolving the coherence/calibration trade-off. A full theoretical treatment of distributionally-robust coherent risk measures, characterising which choices of uncertainty set around the empirical law (Wasserstein balls, φ -divergence balls, or a purpose-built set calibrated directly from Bassi, Embrechts and Kafetzaki's asymptotic quantile-estimation theory (7), rather than from bootstrap resampling) yield an estimator that is *simultaneously* provably coherent (Proposition 6.1) and well-calibrated in the sense of Section 6.5, is left as the natural continuation of this thesis's original contribution.

Liability-driven allocation with an explicit liability process. Section 4.5 draws on Taflin's introduction (6) only as a motivating frame for reading the dynamic risk-budget constraint of Section 4.4 as a solvency capital requirement on an insurer's surplus. Extending the numerical experiments of Chapter 6 to an explicit, possibly random, liability process rather than the pure asset-allocation problem studied there would test the

framework against the application that originally motivated this reading.

Regulatory connections. AVaR_λ at $\lambda = 97.5\%$ is, since the Basel Committee’s Fundamental Review of the Trading Book, the regulatory standard for market-risk capital under the name Expected Shortfall, replacing Value-at-Risk precisely because of the failure of subadditivity established in Proposition 2.3. Connecting the efficient-portfolio theory of this thesis explicitly to regulatory capital constraints rather than to an unconstrained risk budget $\bar{c}$ chosen freely by the investor would be a natural practitioner-facing extension, and would give the discrete-time insurance application of Section 4.5 a market-risk analogue under Basel rather than only a solvency analogue under an insurance capital regime.

Chapitre 8

Conclusion

This thesis set out to establish the state of the art of the theory of convex risk measures and their use in portfolio management, and to formulate and solve the resulting optimal portfolio problem across the mono-period, discrete-time and continuous-time settings, complemented by numerical simulations, as required by the subject. Each part of that programme has been carried out, and is summarised below before a direct answer to the underlying problem, a set of perspectives, and a closing remark on method.

8.1 Summary of contributions

Chapter 2 developed the axiomatic theory of coherent and convex risk measures from first principles, proving the robust representation theorem in full for finite Ω (Theorem 2.2) rather than citing it, exhibiting Value-at-Risk's failure of subadditivity through an explicit, parametrically verified counterexample (Proposition 2.3), and positioning AVaR $_{\lambda}$ as the risk measure used throughout the remainder of the thesis on the strength of its coherence and its variational representation. Chapter 3 formulated the AVaR-efficient portfolio problem as a jointly convex program (Proposition 3.3), established existence on a compact constraint set (Proposition 3.2) after an initial, more ambitious attempt at a universal coercivity bound was checked and found false, and proved the exact coincidence of the AVaR-efficient and Markowitz frontiers under elliptical returns (Theorem 3.4). Chapter 4 extended the theory to discrete multi-period trading, proved that the recursive composition of one-step conditional risk measures is automatically time-consistent (Proposition 4.2), and showed structurally why the naive, static alternative generally is not. Chapter 5 developed the continuous-time analogue for continuously-rebalanced constant-mix strategies in the Black-Scholes model, decomposing the AVaR-optimal proportion into the classical

Merton myopic proportion minus an explicit, interpretable risk-aversion correction (Theorem 5.3). Chapter 6 implemented and tested every one of these results in a Python package of 45 tests and five executed notebooks, and contributed, beyond the scope of the required references, a nested-bootstrap correction for the statistical estimation risk inherent in optimising a risk measure against a finite sample (the “optimizer’s curse”), verified directly against the axioms of a convex risk measure rather than assumed to satisfy them.

8.2 Answer to the problem

Can an efficient-portfolio theory be built on convex risk measures, in place of variance, across static and dynamic settings, and does it change anything that matters? The answer given by this thesis is qualified but clear. In the mono-period, elliptically-distributed case, the answer is that it changes nothing beyond notation : Theorem 3.4 shows the AVaR-efficient and Markowitz frontiers coincide exactly, portfolio for portfolio, for every confidence level. This is not a limitation of the convex-risk-measure approach ; it is the correct and reassuring benchmark against which any more general claim must be checked, and it was checked numerically to within a few percent of Monte Carlo noise (Chapter 6, Notebook 1) before any divergence was claimed. Once returns leave the elliptical family (the empirically realistic case), the answer changes : a single skewed asset was enough to move the AVaR-efficient and Markowitz portfolio weights apart by 35% on average, a divergence variance is structurally unable to see. In discrete time, the theory further shows that *how* a dynamic risk assessment is built, not only *which* risk measure is chosen, matters : two seemingly reasonable constructions (recompute AVaR_λ at each date directly, or compose one-step conditional assessments recursively) can disagree by a large margin for the same terminal position, and only the second is guaranteed to be time-consistent. In continuous time, the theory shows that a convex risk constraint does not discard classical portfolio theory but refines it : the AVaR-optimal constant-mix proportion is the Merton myopic proportion minus an explicit, checkable correction, recovering the risk-neutral myopic policy exactly as the confidence level relaxes to one. Finally, the numerical work shows that the theory’s own machinery, the robust representation of Chapter 2, can be turned back on a distinctly practical problem : estimation risk in data-driven portfolio optimisation, and that doing so honestly means reporting a genuine trade-off (a provably coherent correction that overcorrects, and a well-calibrated one that is not provably

coherent) rather than a single flattering number.

8.3 Perspectives

Four extensions follow directly from the limitations discussed in Chapter 7 and are not repeated in full here : (i) redeveloping the theory for a general convex risk measure rather than AVaR_λ specifically ; (ii) constructing a genuinely dynamic, time-consistent continuous-time risk process via backward stochastic differential equations, the single most significant gap left open by the required references ; (iii) resolving, rather than merely documenting, the trade-off between provable coherence and empirical calibration uncovered in the nested-bootstrap experiment ; and (iv) connecting the discrete-time insurance application and the continuous-time theory to explicit regulatory constraints (Solvency II capital requirements and the Basel Committee’s Expected Shortfall standard, which is exactly $\text{AVaR}_{0.975}$, adopted for the reason established in Proposition 2.3).

8.4 A closing remark on method

Beyond its specific results, this thesis has tried to demonstrate a particular way of working : deriving a piece of theory, checking it independently wherever a check is available, and treating any disagreement as a signal to be chased down rather than smoothed over. A conjectured coercivity bound was checked against an explicit counterexample and discarded before it could be relied upon (Section 3.2) ; a claim that a bootstrap-robustified risk measure had improved on the naive estimator was checked against three variants rather than one, and the variant with the cleanest theoretical guarantee turned out to be the one that performed worst in practice ; a claim that an additive bias correction preserved coherence was checked against all four axioms directly and found to be false in exactly one of them. None of these corrections changed the overall conclusions of the thesis, but each of them would have been a quiet, undetected error had it not been checked. That discipline is, if anything, the most transferable outcome of this particular piece of work, and the one most worth carrying into whatever follows it.

Bibliographie

- [1] H. Markowitz, “Portfolio selection,” *The Journal of Finance*, vol. 7, no. 1, pp. 77–91, 1952.
- [2] P. Artzner, F. Delbaen, J.-M. Eber, and D. Heath, “Coherent measures of risk,” *Mathematical Finance*, vol. 9, no. 3, pp. 203–228, 1999.
- [3] H. Föllmer and A. Schied, “Convex measures of risk and trading constraints,” *Finance and Stochastics*, vol. 6, no. 4, pp. 429–447, 2002.
- [4] H. Föllmer and A. Schied, *Stochastic Finance : An Introduction in Discrete Time*, 2nd ed., ser. de Gruyter Studies in Mathematics. Berlin : de Gruyter, 2004, vol. 27.
- [5] S. R. Pliska, *Introduction to Mathematical Finance : Discrete Time Models*. Malden, Massachusetts : Blackwell Publishers, 1998.
- [6] E. Taflin, “Equity allocation and portfolio selection in insurance,” *Insurance : Mathematics and Economics*, vol. 27, pp. 65–81, 2000.
- [7] F. Bassi, P. Embrechts, and M. Kafetzaki, “A survival kit for quantile estimation,” Department of Mathematics, ETH Zürich, Tech. Rep., 1997.
- [8] W. V. Harlow, “Asset allocation in a downside-risk framework,” *Financial Analysts Journal*, vol. 47, no. 5, pp. 28–40, Sept-Oct 1991.

Annexe

1 Notation

This appendix collects, in one place, the notation used throughout the thesis. Symbols are grouped by the chapter in which they are first introduced ; a symbol reused later with the same meaning is not repeated.

Chapter 2 risk measures

$(\Omega, \mathcal{F}, P)$	measurable space and reference (physical) probability measure
$\mathcal{X}$	space of bounded financial positions (profit-and-loss)
ρ	a monetary risk measure, $\rho : \mathcal{X} \rightarrow R$
$\mathcal{A}_\rho$	acceptance set of ρ , $\{X : \rho(X) \leq 0\}$
$\alpha_{\min}(Q)$	minimal penalty function of a probability measure Q
$\mathcal{M}_1(\Omega)$	set of all probability measures on Ω
$\mathcal{Q}$	set of scenarios (coherent case), $\alpha_{\min} \in \{0, \infty\}$
$\text{VaR}_\lambda(X)$	Value-at-Risk of X at level λ
$\text{AVaR}_\lambda(X)$	Average Value-at-Risk of X at level λ
$\mathcal{Q}_\lambda$	$\{Q \ll P : dQ/dP \leq 1/\lambda\}$, the AVaR scenario set
φ, Φ	standard normal density and distribution function
$\kappa(\lambda)$	$\varphi(\Phi^{-1}(\lambda))/\lambda$, the Gaussian AVaR constant
$\text{LPM}_n(\tau, X)$	lower partial moment of order n relative to target τ (Harlow)

Chapter 3 mono-period

r_f	riskless rate
$R = (R^1, \dots, R^d)$	risky asset return vector
$\pi \in \Theta \subseteq R^d$	portfolio weights (excess-return exposures)
x_0	initial wealth
Y_π	excess position, $x_0 \pi^\top (R - r_f \mathbf{1})$
μ, Σ	expected excess return vector, covariance matrix of R
$\bar{c}, \bar{\mu}$	risk budget (P1) / target excess return (P2)
$F(\pi, c)$	joint objective of the variational reformulation (eq. 3.13)
p_s, r_s	scenario probability and excess return, finite-scenario LP

Chapter 4 discrete time

$(\mathcal{F}_t)_{t=0}^T$	filtration, discrete dates $t = 0, \dots, T$
ρ_t	conditional risk measure at date t
$\rho_t^{(1)}$	one-step conditional risk measure, date t to $t + 1$
θ_t	predictable trading strategy
$S = (S_t)$	discounted risky asset price process

$X_t^{x_0, \theta}$	self-financing wealth process
Y_θ	terminal excess position, $X_T^{x_0, \theta} - x_0$

Chapter 5 continuous time

W_t	standard Brownian motion
μ, σ, r	drift, volatility of S , riskless rate (continuous time)
π_t	(possibly constant) fraction of wealth in the risky asset
X_T^π	terminal wealth under strategy π
R_T^π	terminal log-return, $\ln(X_T^\pi/x_0)$
m_π, σ_π	mean and standard deviation of R_T^π (constant-mix)
$\pi^*(\lambda, T)$	AVaR-optimal constant-mix proportion

Chapter 6 numerics

B	number of bootstrap resamples
$\hat{\pi}, \hat{\pi}_b$	sample-optimised portfolio ; portfolio re-optimised on resample b
$\rho_{\text{sup}}, \rho_q, \rho_{\text{nested}}$	bootstrap-sup, bootstrap-quantile, nested-bootstrap estimators
$\hat{\text{bias}}$	nested-bootstrap selection-bias correction (Section 6.5)

2 Two fully worked examples

This appendix works out, by hand and with clean numbers, two computations flagged in the main text as “exigible in the viva” : the recursive time-consistency construction of Chapter 4, and the mono-period AVaR-efficient portfolio of Chapter 3. Both are also implemented and tested in the accompanying code (Appendix 3), so the two routes can be compared directly.

2.1 A two-period tree : naive versus recursively composed AVaR

Take the two-period tree used to motivate Remark 4.2 : at $t = 1$, “Up” occurs with probability $1/2$, leading at $t = 2$ to two equally likely outcomes $\{+10, -100\}$; “Down” occurs with probability $1/2$ and is deterministic, leading to 0. The one-step confidence level is $\lambda_1 = 1/2$ at both dates ; the naive static confidence level is $\lambda_0 = 1/4$.

Step 1 : the terminal distribution. The four leaves, in the order (Up-Up, Up-Down, Down-Up, Down-Down), have payoffs $(10, -100, 0, 0)$ each with unconditional probability $1/4$.

Step 2 : the naive static AVaR_{1/4}. Sorting the outcomes ascending, $-100(1/4), 0(1/4), 0(1/4), 10(1/4)$. The event $\{X < -m\}$ has probability

$$P(m) = \begin{cases} 1 & m < -10 \\ 3/4 & -10 \leq m < 0 \\ 1/4 & 0 \leq m < 100 \\ 0 & m \geq 100, \end{cases}$$

obtained by checking, for each range of m , which of the four sorted outcomes satisfy $X < -m$. The infimum m with $P(m) \leq 1/4$ is $m = 0$, so $\text{VaR}_{1/4}(X) = 0$; for every

$s < 1/4$ strictly, $P(m) \leq s$ requires $m \geq 100$, so $\text{VaR}_s(X) = 100$ on the open interval $(0, 1/4)$. Averaging,

$$\text{AVaR}_{1/4}(X) = 4 \int_0^{1/4} \text{VaR}_s(X) ds = 4 \times 100 \times \frac{1}{4} = 100.$$

Step 3 : the recursively composed ρ_0 . At $t = 1$, in the Up state, the conditional distribution is $\{10(1/2), -100(1/2)\}$; by the same step-function argument, $\text{VaR}_{1/2}$ of this distribution is -10 but $\text{VaR}_s = 100$ for every $s \in (0, 1/2)$, so $\rho_1(X)|_{\text{Up}} = \text{AVaR}_{1/2}(\{10, -100\}) = 100$. In the Down state, the position is deterministically 0, so $\rho_1(X)|_{\text{Down}} = 0$. Composing one more step, $\rho_0(X) = \text{AVaR}_{1/2}(\{-\rho_1(X)|_{\text{Up}}, -\rho_1(X)|_{\text{Down}}\}) = \text{AVaR}_{1/2}(\{-100, 0\})$, and the same computation gives $\text{AVaR}_{1/2}(\{-100(1/2), 0(1/2)\}) = 100$.

Conclusion. Both conventions give exactly 100 on this particular tree *not* because the two are equal in general (Chapter 4 shows they are not), but because a single dominant worst-case path (the Up-Down leaf, reached with probability $1/4$ unconditionally and with one-step probability $1/2$ at each of the two steps that lead to it) happens to drive both computations to the same number here. This is exactly the coincidence that made a hand-constructed genuine counterexample elusive while preparing Chapter 4, and why the divergence reported there (Section 6.4) was instead obtained by random search over many trees rather than by further hand construction.

2.2 A four-scenario AVaR-efficient portfolio

Take two risky assets and four equally likely scenarios ($p_s = 1/4$), with excess returns

$$r_1 = (0.20, -0.10), \quad r_2 = (-0.10, 0.20), \quad r_3 = (0.05, 0.05), \quad r_4 = (-0.05, -0.05),$$

initial wealth $x_0 = 1$, confidence level $\lambda = 1/4$, and target excess return $\bar{\mu} = 0.05$.

Step 1 : symmetry. By inspection, r_1 and r_2 are permutations of one another, and r_3, r_4 are symmetric about zero along the diagonal $\pi^1 = \pi^2$. This suggests searching for the optimum along the diagonal $\pi = (\pi_c, \pi_c)$.

Step 2 : reduce to a one-dimensional problem. Along the diagonal, the four scenario excess payoffs $Y_\pi(\omega_s) = \pi_c(r_s^1 + r_s^2)$ are $Y_\pi(\omega_1) = 0.10\pi_c$, $Y_\pi(\omega_2) = 0.10\pi_c$, $Y_\pi(\omega_3) = 0.10\pi_c$, $Y_\pi(\omega_4) = -0.10\pi_c$. The expected excess return is $E[Y_\pi] = \frac{1}{4}(0.10\pi_c \times 3 - 0.10\pi_c) = 0.05\pi_c$, so the target $\bar{\mu} = 0.05$ binds exactly at $\pi_c = 1$, giving $\pi^* = (1, 1)$ along this diagonal.

Step 3 : the AVaR of this candidate. At $\pi_c = 1$, the four payoffs are $(0.10, 0.10, 0.10, -0.10)$, each with probability $1/4$; the single worst outcome -0.10 has probability exactly $1/4 = \lambda$, so by the same mechanism as Appendix 2.1, $\text{AVaR}_{1/4}(Y_{\pi^*}) = -(-0.10) = 0.10$.

Step 4 : verifying optimality off the diagonal. Any perturbation $\pi = (1 + \varepsilon, 1 - \varepsilon)$ leaves the expected return at 0.05 unchanged only to first order in the symmetric scenarios ω_1, ω_2 (their payoffs move in opposite directions and average out) but strictly increases the magnitude of the worst-case payoff at ω_4 for $\varepsilon \neq 0$ (since r_4 is symmetric, $Y_\pi(\omega_4) = -0.05(1 + \varepsilon) - 0.05(1 - \varepsilon) = -0.10$ is in fact unchanged at ω_4 , but ω_1, ω_2 payoffs become

$0.20(1 + \varepsilon) - 0.10(1 - \varepsilon)$ and $-0.10(1 + \varepsilon) + 0.20(1 - \varepsilon)$ respectively, one of which strictly decreases below 0.10 for any $\varepsilon \neq 0$, replacing the single worst scenario with a strictly worse pair and strictly increasing $\text{AVaR}_{1/4}$; the diagonal candidate is confirmed optimal by this direct perturbation argument, consistently with the LP solution reported for exactly this example in the accompanying regression test.

3 Reproducibility and traceability

All numerical results reported in this thesis are produced by the accompanying code repository (package `convexrisk`), organised so that the main theoretical claims of Chapters 3–6 each correspond to a specific test or notebook.

3.1 Installation

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
python -m pytest tests/ -v
```

As of the version underlying this thesis, this runs 45 tests.

3.2 Running the notebooks

```
jupyter nbconvert to notebook execute inplace \
    notebooks/01_mono_period_avar_frontier.ipynb
```

and similarly for notebooks 02 through 05; each is self-contained and depends only on the installed `convexrisk` package and standard scientific Python libraries (`numpy`, `scipy`, `matplotlib`).

3.3 Correspondence between thesis claims and code

Claim	Test / notebook
Robust representation (Thm. 2.2)	proved directly in the text ; not separately coded
VaR fails subadditivity (Prop. 2.3)	Appendix 2.1-style hand check ; <code>test_risk_measures.py</code>
AVaR of a discrete distribution	<code>test_discrete_avar_exact.py</code> , matched against three hand-derived examples
Existence on a compact Θ (Prop. 3.2)	<code>test_mono_period.py</code>
Gaussian coincidence with Markowitz (Thm. 3.4)	<code>test_mono_period.py</code> ; Notebook 1
Divergence under skewness (Remark 3.4)	Notebook 1 (35% mean weight error)
Recursive composition is time-consistent (Prop. 4.2)	<code>test_discrete_time.py</code> , hand example of Appendix 2.1
Naive vs. composed divergence (Remark 4.2)	<code>search_for_divergence</code> ; Notebook 2 (gap ≈ 2.8)
AVaR of continuously-rebalanced log-return (Prop. 5.3)	<code>test_continuous_time.py</code> ; Notebook 3 (max rel. error 0.24%)
AVaR-optimal constant-mix (Thm. 5.3)	<code>test_continuous_time.py</code> ; Notebook 3
Harlow LPM vs. AVaR vs. Markowitz (Section 2.6)	<code>test_downside_risk.py</code> ; Notebook 4
Sup of coherent measures is coherent (Prop. 6.1)	<code>test_robust.py</code> , all four axioms checked directly
Bootstrap-sup overcorrects	<code>optimizer_curse_experiment</code> ; Notebook 5
Nested bootstrap is convex, not coherent	<code>test_nested_bootstrap_is_convex_but_not_coherent</code> ; Notebook 5